



Orion-MC

Manual Air Mixing

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1 Introduction

The Orion-MC is the communication interface between a horticulture computer (the main controller) and a variety of systems in the horticultural greenhouse (air mixing units).

The horticulture computer communicates with the Orion-MC using separate control signals.

The Orion-MC communicates with the IO modules via a CAN Local data bus. The Orion-MC receives control commands from the main controller. The Orion-MC interprets these control commands in terms of commands for the relevant IO module. The Orion-MC also sends the main controller feedback on the position adopted by the controlled element.

The Orion-MC also offers a number of additional functions. The *fine adjustment* option enables you to make fine adjustments to the heaters of the air mixing unit, so that the air each air mixing unit blows into the greenhouse has the same temperature.



Safety instructions and warnings

This product should be installed by an approved, certified installer in accordance with the relevant standards (BS, ANSI, ISO, for example).

The warranty is void if this product is installed in a switchbox other than by Hotraco Horti BV. The installation incorporating this product should be equipped with an appropriate safety system that is inspected at periodic intervals (at least once a day) to verify that it is working correctly.

Please note that when the product is included in a climate control system, the oxygen content in the climate controlled area must be guaranteed at all times, including during an emergency. To guarantee the oxygen content we recommend an emergency system, which must be periodically inspected for proper operation.

An emergency system can consist of:

- an emergency power generator;
- emergency opening systems;
- provision for manual operation.

We wish to draw your attention to the following:

- The Orion-MC computer is an electronic product. Consequently, you always need to take into consideration a possible malfunction.

The alarm contacts of the various computers should be connected to a central alarm system.

Furthermore, Hotraco Horti BV recommends the installation of an additional independent alarm system (such as a min./max. thermostat).

Although Hotraco Horti BV has made every effort to ensure that adequate alarms are generated in the event of emergencies, Hotraco Horti BV cannot guarantee that all possible situations have been covered.

- We recommend that the computer remains powered up whenever possible. It is preferable to keep the computer powered up when not in use, since this will prevent condensation.
- Do not clean the computer with running water: the computer is splashproof, but not waterproof!
- This appliance is covered by the warranty and liability regulations laid down in documents including the Hotraco Horti general terms of sale governing the contract concluded for the delivery of this appliance.
- Use other than described in this manual is not permitted without express consent from Hotraco Horti BV.
- The Orion-MC is under no circumstances a safety component.
- The Orion-MC may only be operated and adjusted by persons who are familiar with the operating instructions provided in this manual.
- The person who operates or adjusts the Orion-MC must be aware of all potential dangers of the systems that the Orion-MC controls.
- A defective Orion-MC is hazardous! In the event of a fault, the systems in which the Orion-MC is used are to be taken out of service until the Orion-MC has been repaired.

- It is not permitted to connect external devices that do not have a SELV or Class II power supply (double insulated).
- Hotraco Horti wishes to point out that Machinery Directive 2006/42/EC is applicable when the Orion-MC is used to control moving systems.
- Due to the nature of the Orion-MC controllers, a controlled system can start at any time. Hotraco Horti would like to point out that this may result in hazardous situations and danger from moving parts when such a system is in operation. Therefore, people are not permitted to be in the immediate vicinity of moving parts. The moving parts must be fitted with guards and/or safety devices.
- Motors and actuators that are directly controlled via the Orion-MC may never be connected without a suitable motor protector and a lockable isolating switch.

End of life (WEEE)

The crossed-out wheeled bin indicates a product that, at the end of its life cycle, must be collected separately (Directive 2012/19/EU on waste electrical and electronic equipment - WEEE). Separate collection is a precondition to ensure specific treatment and recycling of WEEE to get the chosen level of protection of human health and the environment in the Union. The black bar below the crossed-out wheeled bin indicates that the product was put on the market on or after August 13, 2005



The European Union has systems in place for the separate collection of waste electrical and electronic products and batteries. Thus, return waste products, batteries and accumulators to the appropriate local authorities, or a local collection point/recycler. When you do not dispose of the product correctly, you may be fined.

2 Operation

2.1 Operating keys

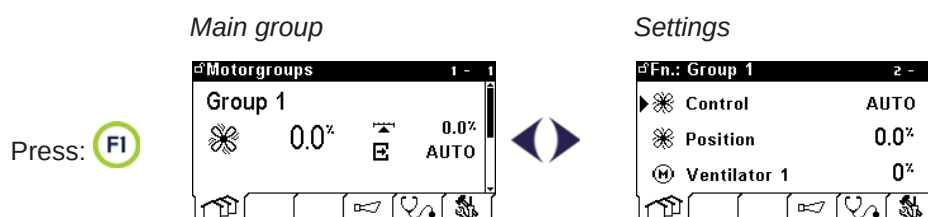
The Orion-MC is equipped with a number of operating keys. These keys can be classified into the following groups:

- Function keys:
-  F1 Time and date
 -  F2 Groups, air mixing units
 -  F3 No function
 -  F4 Alarms
 -  F5 Diagnostics
 -  F6 Installation procedure

- Navigation keys:
-  Move cursor
 -  Increase value / Previous line
 -  Decrease value / Next line
 -  Move cursor / Correction of numeric entry
 -  OK Confirm value
 -  Next motor or motor group
 -  Previous motor or motor group

2.2 The readout

The display can show information in several ways. The section describes the main ways.



The *main group readout* displays the current data and/or the status of the controlled element. These data relate to the measured and calculated positions and can be fetched by pressing the **F1** key.

The data in the *main group readout* cannot be changed. You can change settings relating to the data displayed in the main group by pressing the **▶** key, which opens the settings readout. Press the **◀** key to return to the *main group readout*.

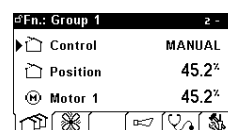
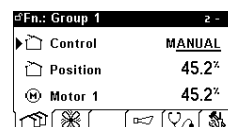
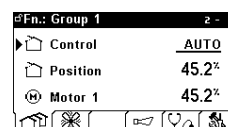
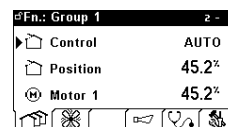
A list of the possible main groups and settings is enclosed in appendix 1. Data relating to controlled elements that have not been activated are skipped automatically and, consequently, cannot be fetched.

2.3 Entering and adjusting settings

Press the (**F1** or **F2**) function key to display the *main group screen*. Pressing the **▶** key opens the *settings screen*. You can fetch the required setting using the **▲** and **▼** keys.

Example Changing a group to manual operation mode:

1. Press **F1** or **F2**: the main group is displayed:
2. Press **▶**: the settings screen is displayed.
3. Press **▶**: the cursor is moved to the setting.
4. Press **▲**: the setting is changed.
5. Press **OK**: the changed setting has been confirmed.



2.4 System settings

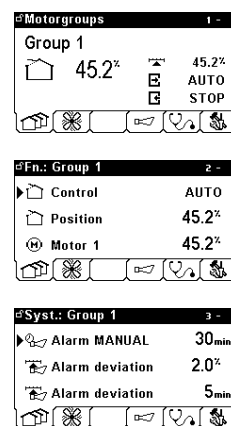
System settings refer to settings relating to the minor changes that can be made to the system. The basic settings have already been configured: the Orion-MC will carry out its control functions satisfactorily without any changes to the system settings.

However, you can change the system settings to tailor the Orion-MC precisely to the equipment that is to be controlled. We recommend that you make any changes in consultation with your installer.

The system settings are grouped by controlled element. The system settings for the relevant controlled element can be fetched and adjusted by opening the *settings* screen and pressing the **F6** key. The system settings that are not of relevance are not displayed.

Example System settings of a group:

1. Press **F1** or **F2**: the main group is displayed:
2. Press **▶**: the settings screen is displayed.
3. Press **F6**: the system settings are displayed. System settings are adjusted using the procedure used for the adjustment of settings, see chapter 2.3.

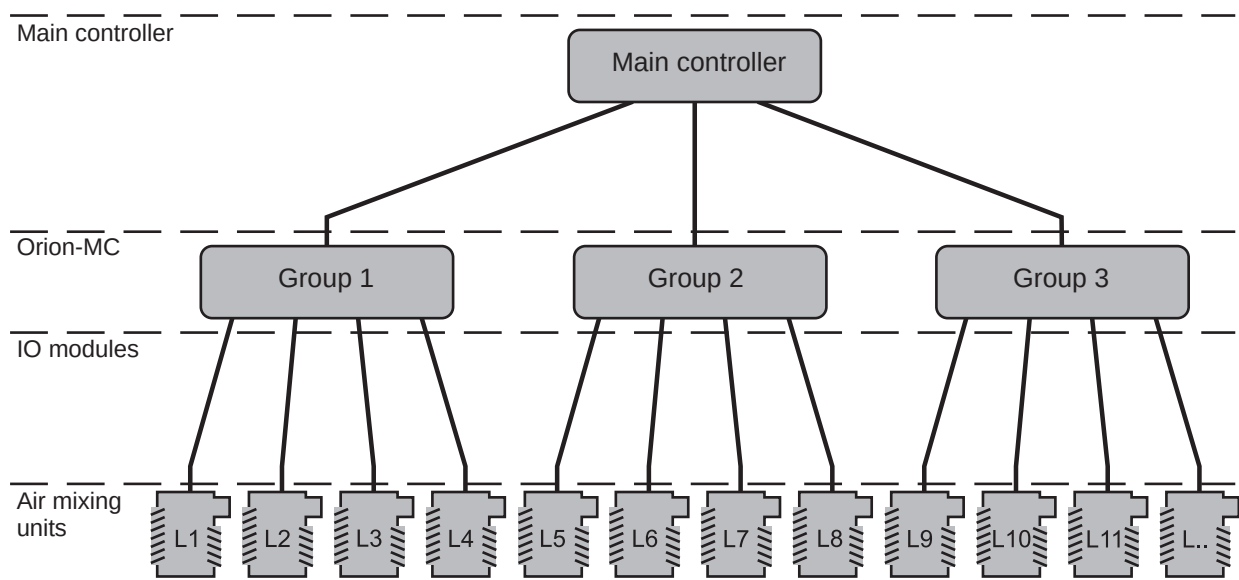


3 The Orion-MC's function

3.1 Control of the air mixing units

The building is equipped with a number of air mixing units (L1, L2 , etc., in the following diagram). These air mixing units are connected to Hotraco Industrial IO modules. The IO modules are in turn connected to the Orion-MC. More information is enclosed in Annex 2.5 .

The air mixing units are clustered into air mixing unit groups in the Orion-MC. Each air mixing group consists of one or more air mixing units that are, in principle, controlled simultaneously and all adopt the same position. The main controller controls the air mixing unit groups and the Orion-MC ensures that the air mixing units adopt the correct position.



You can use the Orion-MC to control all the air mixing units in a group or, if you so wish, to control a single air mixing unit.

3.2 Virtual control

The horticulture computer serves as the controller and determines the required action for each group. The horticulture computer sends the relevant action to the Orion-MC. The Orion-MC simulates this action and sends the horticulture computer feedback on the simulated action (= virtual control). The horticulture computer interprets this feedback as confirmation that the required action has been carried out. At the same time the Orion-MC controls the IO modules by sending commands to ensure that the group carries out the required action. The relevant units send feedback on the positions to the Orion-MC. The action is stopped once the required position has been reached or an alarm is generated if the required position is not reached.

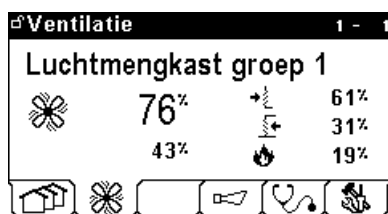
4 Air mixing unit settings

4.1 Air mixing units with inner and outer flaps

More information about air mixing units with inner and outer flaps is enclosed in Appendix 2.2.

4.1.1 Group

Press **F2** to display the main **group** screen. Navigate through the groups with ▲▼.

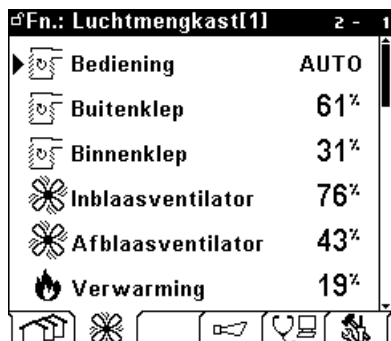


This screen displays the status of a **group** with air mixing units.

- 76%** = The required ventilation position of all air blast fans in the group.
- = The required position of all the outer flaps in the group.
- = The required position of all the inner flaps in the group.
- = (Option) the required position of all the heaters in the group.
- 43%** = (Option) the required ventilation position of all air exhaust fans in the group.

4.1.2 Group settings

Press  in the main group to display the settings screen. Navigate through the groups with .



This screen displays the settings of the **group**.



= *Operation*: Change the status of the group:

AUTO The group is controlled automatically; the main controller determines the required position of the air mixing units and the Orion-MC ensures that the air mixing units are set at the correct position.

HAND The group is controlled manually; the group can be controlled via the following settings from the Orion-MC.

OFF The group is switched off.



= *Outer flap*: The required position of all the outer flaps in the group. When the *Operation* of the group is set to *HAND* then you can set the position here.



= *Inner flap*: The required position of all the inner flaps in the group. When the *Operation* of the group is set to *HAND* then you can set the position here.



= *Air blast fan*: The required position of all air blast fans in the group. When the *Operation* of the group is set to *HAND* then you can set the ventilation position here.



= *Air exhaust fan*: (option) the required position of all air exhaust fans in the group. When the *Operation* of the group is set to *HAND* then you can set the ventilation position here.



= *Heating*: (option) the required position of all heaters in the group. When the *Operation* of the group is set to *HAND* then you can set the heating here.

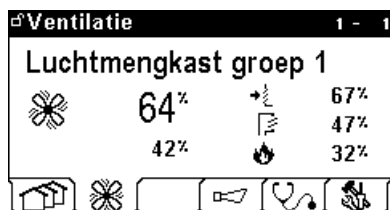
Note You can operate each air mixing unit by hand using the diagnostics function, see chapter 6.

4.2 Air mixing units with recirculation

More information about air mixing units with recirculation flap is enclosed in Appendix 2.2.

4.2.1 Group

Press  to display the main **group** screen. Navigate through the groups with  .



This screen displays the status of a **group** with air mixing units.

- 64% = The required ventilation position of all air blast fans in the group.
- = The required position of all the recirculation flaps in the group.
- = (Option) the required position of all the upper flaps in the group.
- = (Option) the required position of all the heaters in the group.
- 42% = (Option) the required ventilation position of all air exhaust fans in the group.

4.2.2 Group settings

Press  in the main group to display the settings screen. Navigate through the groups with  .



This screen displays the settings of the **group**.

-  = **Operation:** Change the status of the group:
 - AUTO** The group is controlled automatically; the main controller determines the required position of the air mixing units and the Orion-MC ensures that the air mixing units are set at the correct position.
 - HAND** The group is controlled manually; the group can be controlled via the following settings from the Orion-MC.
 - OFF** The group is switched off.
-  = **Recirculation flap:** The required position of all the recirculation flaps in the group. When the *Operation* of the group is set to *HAND* then you can set the position here.
-  = **Upper flap:** (Option) the required position of all the upper flaps in the group. When the *Operation* of the group is set to *HAND* then you can set the position here.
-  = **Air blast fan:** The required position of all air blast fans in the group. When the *Operation* of the group is set to *HAND* then you can set the ventilation position here.
-  = **Air exhaust fan:** (option) the required position of all air exhaust fans in the group. When the *Operation* of the group is set to *HAND* then you can set the ventilation position here.
-  = **Heating:** (option) the required position of all heaters in the group. When the *Operation* of the group is set to *HAND* then you can set the heating here.

Note You can operate each air mixing unit by hand using the diagnostics function, see chapter 6.

4.3 Group system settings

In the **group** settings screen, press  to get the system settings. Navigate through the groups of the air mixing unit with .

SYS Description

-  Alarm, hand
-  Alarm variance (min. variance before alarm is generated)
-  Alarm variance (min. wait time before alarm is generated)
-  Air exhaust fan on
-  Air exhaust fan minimum
-  Air exhaust fan maximum
-  Hysteresis

Default	User
30 min	min
5%	%
10 min	min
20%	%
20%	%
80%	%
2%	%

SYS Description

	Step size
	Cycle time
	Hysteresis
	Bandwidth

Default	User
5%	%
30 s	s
0.5 °C	°C
1.0 °C	°C

Alarm, hand

When the **group** with air mixing units is still set to hand operation mode on the Orion-MC after the set time then the group's alarm contact is de-energised. This function can be disabled by setting *alarm, hand* at 0 minutes.

Alarm variance

You can configure the Orion-MC to generate an alarm in the event that an air mixing unit is not at the required position after the set *wait time*.

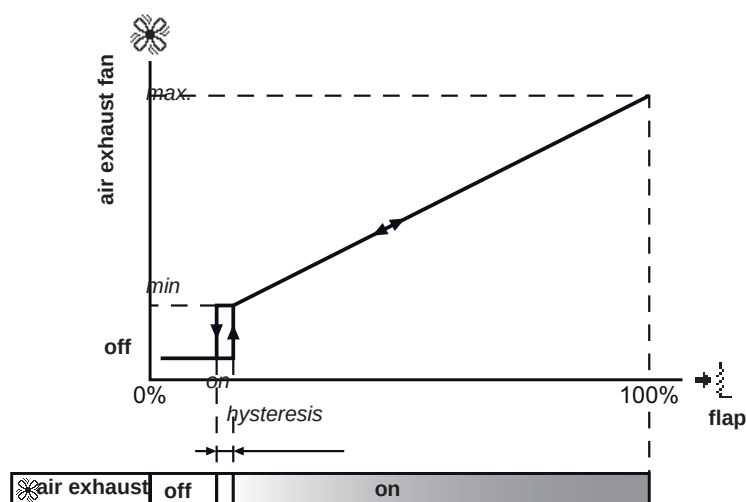
Note You can fetch the position of the individual air mixing units using the diagnostics function, see chapter 6.

Linked air exhaust fan (option)

These settings are applicable when the air exhaust fan is linked to the position of the exterior or recirculation . (This can be configured during the installation procedure, see chapter 7.3).

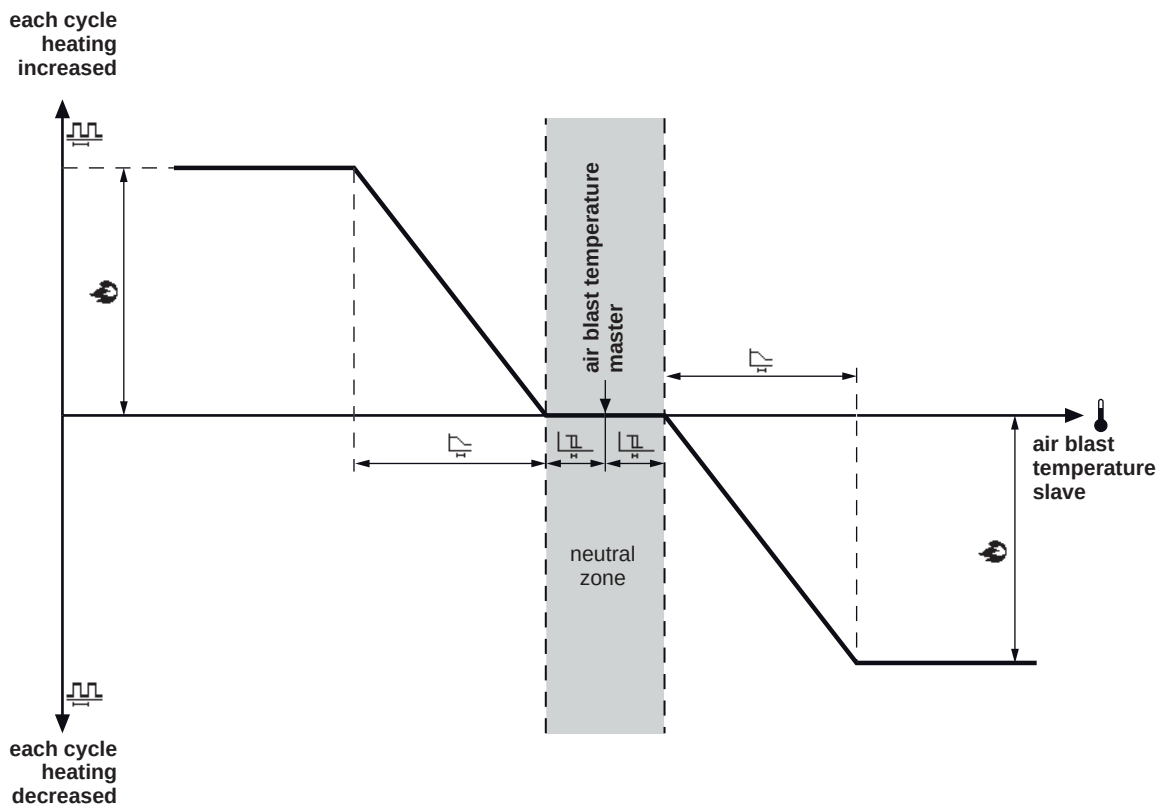
The air exhaust fan is switched on when the position is greater than the *air exhaust fan on* setting. The air exhaust fan is switched off when the position is less than the *air exhaust fan on* setting **minus** *hysteresis*.

The ventilation position on switching the air exhaust fan on is equal to the *air exhaust fan minimum* setting. When the is opened further then the speed of the air exhaust fan will increase. When the is opened to the maximum position then the ventilation position is equal to the *air exhaust fan maximum* setting.



Fine adjustment (option)

The fine adjustment option is available only when the air mixing units are equipped with heating and a temperature sensor. This option can be enabled during the installation procedure (see chapter 7.3). The main controller controls the heating of the group on the basis of temperature measurements of one air mixing unit. This is the master. The fine adjustment corrects the heaters of the other air mixing units on the basis of the *air blast temperature* of the master.



When the *air blast temperature slave* is in the neutral zone (*air blast temperature master* $\pm$ hysteresis (L_{H})) then no correction will be made.

When the *air blast temperature slave* is **lower** than the *air blast temperature master minus hysteresis* (L_{H}) then the heating in the air mixing unit is increased until the temperature returns to the neutral zone. The heating is increased in increments. The heating is increased by a given step size once every *cycle time* (L_{CT}). The minimum step size is 0% and the maximum step size can be set under *step size* (L_{SS}). The maximum step size is reached when the *air blast temperature slave* is lower than the *air blast temperature master minus hysteresis* (L_{H}) *bandwidth* (L_{BW}).

When the *air blast temperature slave* is **higher** than the *air blast temperature master minus hysteresis* (L_{H}) then the heating in the air mixing unit is decreased until the temperature returns to the neutral zone. The procedure is the same as that for an increase in the heating.

5 Alarms

5.1 Alarm messages

5.1.1 Alarm types

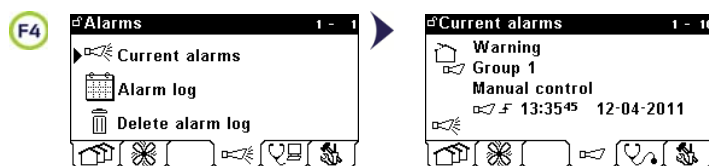
The Orion-MC uses the following types of alarms (F4).

-  = Soft alarm. The alarm contacts on the IO boards remain energised.
-  = Several soft alarms.
-  = Hard alarm. The alarm contacts on the IO boards are de-energised.
-  = Several alarms, including at least one hard alarm.

Each IO board has a separate alarm contact that is de-energised in the event of a communication failure, power failure and a **hard alarm**. The alarm contact activates, for example, a claxon, revolving lamp and/or telephone dialler. Your installer will configure the necessary settings for additional alarm contacts during the installation procedure. The following options are available:

- Each group has a separate alarm contact that is de-energised in the event of a **hard alarm** relating to the relevant group.
- A separate alarm contact that is de-energised on each **soft alarm**.

5.1.2 Alarm display



5.1.3 Resolve the alarms

Use the following procedure in the event of an alarm:

1. Press F4 to view which alarm has been generated. When there are several alarms then you can use the ▼▲ keys to browse through the alarms.

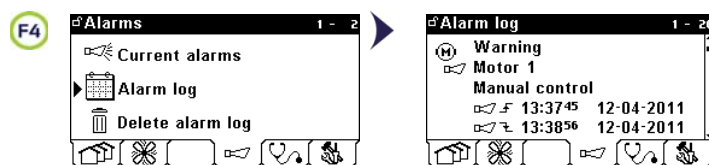
- Most alarms can be converted into another type of alarm. You do so by pressing **OK** for the alarm you wish to convert and then using the $\nabla/\blacktriangle$ keys to change the type of alarm.

-  = **Soft alarm:** The alarm has yet to be resolved, but the alarm contacts on the IO boards are re-energised. The alarm contact of the group remains de-energised. The separate alarm contact for soft alarm is de-energised.
-  = **Suppressed soft alarm:** The alarm has not yet been resolved but will no longer be generated. The alarm contact of the group remains de-energised. When the alarm contact for soft alarm is de-energised then this contact is re-energised.
-  = **Hard alarm:** When the alarm contact for the group has been de-energised then this remains de-energised. When the alarm contact for soft alarm is de-energised then this contact is re-energised. The alarm contacts on the IO boards are de-energised.
-  = **Deleting an alarm because it has been resolved:** If the alarm has been resolved then you can delete the alarm to restore the control.

- Investigate the cause of the alarm.
- Solve the problem in a **safe manner**.
- Many alarm messages will be cancelled automatically once the problem has been resolved. When necessary, alarms will be cancelled only once you have deleted the message with the  command. When an alarm cannot be cancelled then there is still a problem.
- Once all alarms have been cancelled *No current Alarms* is displayed in the screen.

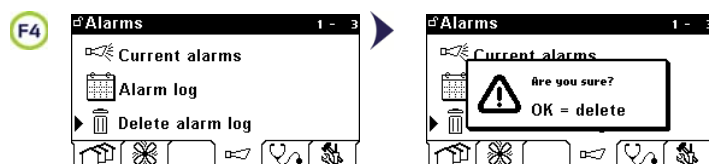
5.1.4 Alarm log

Once an alarm has been resolved then the alarm message is saved in *Alarm log*. The time at which the alarm was resolved is added to the last line. You can use the keys to browse through the alarm messages.



5.1.5 Deleting alarms

You can use *Delete alarm log* to delete the alarms.



5.2 Alarm messages from the IO modules

5.3 Orion-MC Alarms

Description	Type	Cause
 Air mixing unit group 'xx' Hand mode.	Alarm/Warning	The group number 'xx' is set to hand-operation mode in the Orion-MC. The alarm contact will be de-energised after a while (see system settings).
 Air mixing unit 'xx' position not reached.	Alarm	The relevant component of the air mixing unit has not reached the required position, see also chapter 4.3.
 Air mixing unit 'xx' General external alarm.	Alarm	Air mixing unit number 'xx' has generated an alarm. Check the air mixing unit.
 Air mixing unit 'xx' Frost protection.	Alarm	The frost protection of air mixing unit number 'xx' has been activated (see also the comment).

Note Frost protection

When the frost protection thermostat switches in then the air mixing unit is brought to the safe state. The outer or recalculation flap is closed, the inner flap is opened to 100%, the fans are switched off and the heating is switched off.

When the frost protection thermostat switches off, then the air mixing unit remains in the safe state until the alarm in the Orion-MC is cancelled:

1. Call the alarm in *active alarms* (see also chapter 5.1.3).
2. Press  once you see the alarm displayed on the screen.
3. Delete the alarm with  .

5.4 System alarms

5.4.1 General alarms

Description	Type	Cause
 Orion switched off	Warning	The software has switched off the computer at the specified time.

The computer has been switched off and gives an alarm. Press any key to suppress the alarm and to convert the alarm into a warning. No alarms from the controllers are shown while the computer is switched off.

Description	Type	Cause
 System restart	Warning	The computer has been restarted at the specified time.
 Settings deleted	Alarm	The settings have been deleted at the specified time (Installation procedure).
 Options deleted	Alarm	The options have been deleted at the specified time (Installation procedure).
 Alarm deleted	Warning	The alarms have been deleted at the specified time (see chapter 5.1).
 Options restored	Alarm	Someone has restored the options.
 Settings restored	Alarm	Someone has restored the settings.
 Options + settings restored	Alarm	Someone has restored the options and settings.
 Options (+ settings) not restored	Alarm	The options and settings that have been restored are not correct.

5.4.2 Communication alarms

Description	Type	Cause
 Board IO-xx-xx [xx] not found.	Alarm	The Orion has no connection with the relevant IO board. Possible causes: • Communication interrupted. • Broken contact. • CAN bus terminator defect.
 No alarm contact.	Alarm	No alarm contact assigned. The IO boards are not configured correctly (Installation procedure).
 Air mixing unit air blast or exhaust fan 'xx' not found.	Alarm	There is a communication problem with the RS485 connection (ebmBus on the IO-07-07 board).

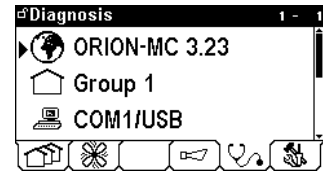
5.4.3 System alarms (internal)

The Orion-MC tests a number of internal functions that are directly applicable to the computer. On detecting an error the computer generates a system alarm displaying one of the following messages. Always warn your installer in the event of a system alarm!

Description	Type	Cause
 Options corrupted	Alarm	System setup no longer correct.
 Settings corrupted	Alarm	System settings are no longer correct.
 Minimum/maximum value corrupted	Warning	The stored minimum and maximum values are no longer correct.
 I ² C error	Alarm	Internal communication problem in the Orion.
 EEPROM error	Alarm	Internal communication problem with the EEPROM settings and options.
 RTC error	Warning	Problem with the real-time clock (program flow).
 TIMER error	Warning	Internal clock is not working correctly.
 HTRAP error	Warning	Internal memory is not working correctly.
 PLL error	Warning	PLL circuit is not working correctly.
 IO board ADC error	Warning	Analog/Digital converter not working properly.

6 Diagnostics

Press the **F5** diagnostics keys to get access to extra functions. You can use these functions to access information about the system elements. The air mixing unit heading gives you access to information about all the available air mixing units. If you so require you can set all the available air mixing units to hand operation mode. The diagnostic functions are displayed by pressing the **▶** key for the relevant element.



You can also fetch information about the various communication lines. Only the installed communication lines are displayed (see chapter 6.3).

You can also use the diagnostics screen to read the data of a SD card or write data to a SD card. More information is given in chapter 6.4

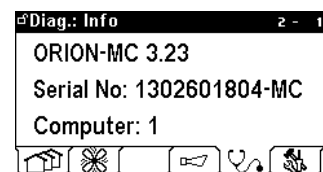
6.1 Software version, serial number and module

Software version

The diagnostics opening screen displays the software's version (in this example Orion-MC 4.18). Since the majority of this computer's functions are determined by the software it is important that the number in front of the decimal point (in this example 4.18) is in agreement with the number of this manual (see the version number on the front page + footer texts).

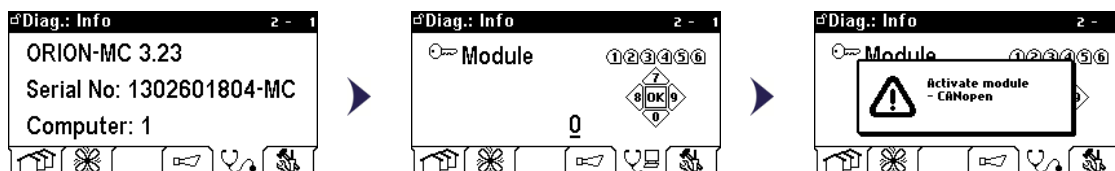
Serial number

Each Orion is assigned a unique serial number. When the cursor is on *Orion-MC*, pressing the **▶** key displays the screen with the serial number. This can be applicable to a software update.



Module

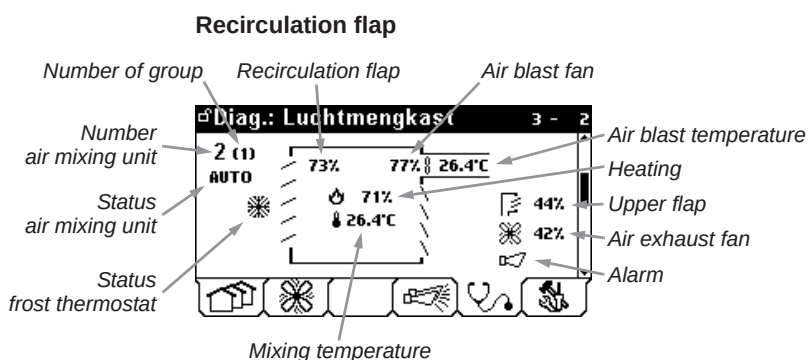
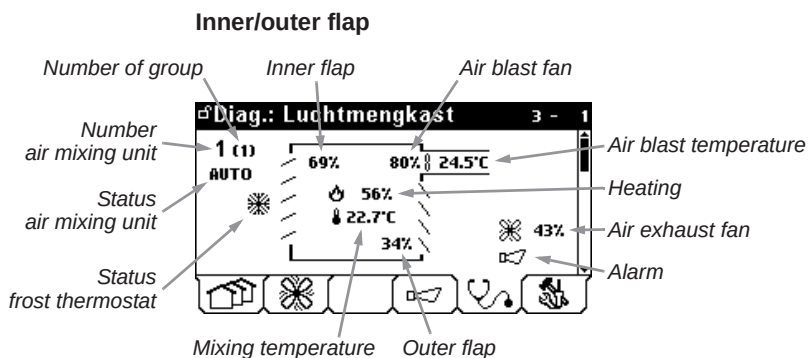
You can change or add a module (for example, CANopen communication) by entering a code under *Module* (C=).



Note You can request the code from your supplier.

6.2 Air mixing units

Press the  key when the cursor is on *Air mixing unit* to get diagnostics information for the connected air mixing units. The air mixing units can be selected with the   keys. Only the installed system components are displayed.



You can adjust the status of the air mixing unit if you so wish.

1. Press the  key to position the cursor in the screen. Navigate through the settings with .
2. Set the cursor to the *status air mixing unit* position.
3. Change the status with the   keys. The following options are available:
 - **Auto** The screen displays the values reported back by the air mixing unit. These are the actual positions.
 - **Hand** The screen displays the values reported back by the air mixing unit. However, you can now adjust the set position of the relevant air mixing unit.
 - Set the cursor on the parameter that must be changed.
 - Change the position with the   keys.
 - Use the  keys to move to the next value in the screen.
 - **Off** The air mixing unit is switched off. All components are controlled to the initial position (this initial position will ultimately be reported back).
4. Press, once the cursor is on *status of air mixing unit*, the  key to exit the screen.

6.3 Communication

6.3.1 COM1/USB and COM2

Diag.: COM1/USB 2 - 1

COM1/USB		38.4kBd
TX	94	449
RX	5	62
TX+RX	99	463

Diag.: COM2 2 - 1

COM2		38.4kBd
TX	114	243
RX	0	0
TX+RX	113	243
AT	☐	
Modem DCD	OK	☐
Modem Init	☐	

Communication information

Send AT command to the Modem

Test DCD signal

Send initialisation string to the Modem

6.3.2 CAN Local (IO boards)

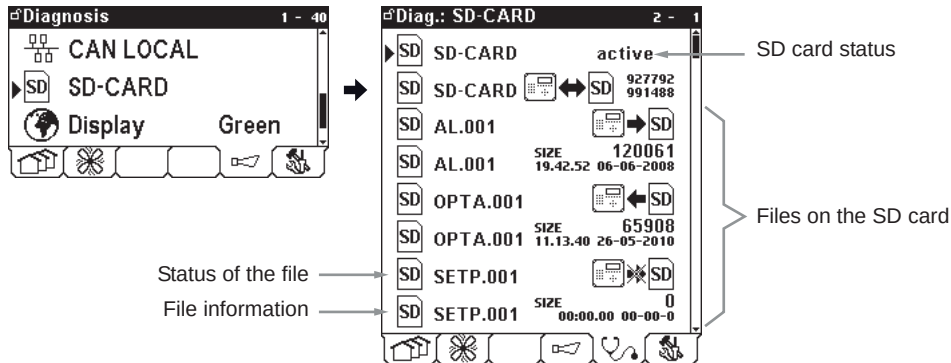
Diag.: CAN LOCAL 2 - 1

TXOK	1	225
RXOK	1	241
TXOK+RXOK	0	459
BOFF	0	
EWRN	0	
Stuff error	0	
Form error	0	
Ack error	0	
Bit1 error	0	
Bit0 busoff error	0	
Bit0 normal error	0	
Crc error	0	

Communication information

6.4 SD card

6.4.1 SD card settings



Changing the SD card status

Press the **▶** key when the cursor is on *SD CARD*. Change the status with the **▲▼** keys. Confirm with the **OK** key. You can assign the following statuses to the SD card:

Remove: The SD card can be removed safely from the Orion-MC.

Ask remove: Data is still being written to the card, the SD card cannot be removed yet.

Current: The SD card can be used to save or restore data.

Changing the status of the file

Press the **▶** key when the cursor is on the file. Change the status with the **▲▼** keys. Confirm with the **OK** key. You can assign the following statuses to the SD card:



No action takes place.



The data are stored on the SD card. With log data the data are supplemented continually and the **➡** symbol is always displayed. With one-off data (such as options) the symbol changes back to **✖** once the data has been saved.



The data are read out from the SD card and send to the Orion (for example, options). Once the read-out has been completed the symbol changes back to **✖**.

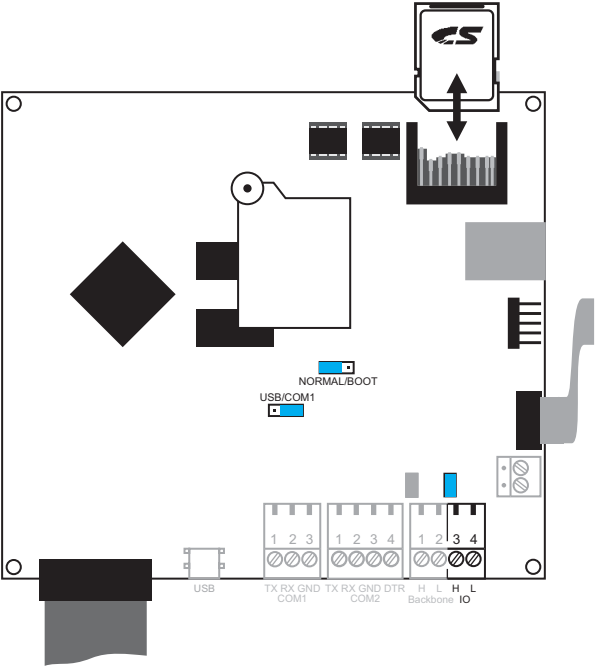
Data on the SD card

Alarm list (AL), options (OPT) and setpoints (SETP).

6.4.2 SD card location

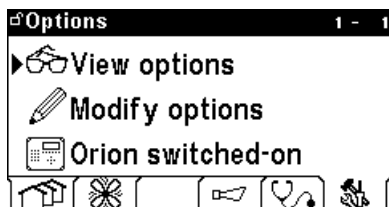
The SD card is in the case of the Orion-MC. Open the case to access the SD card. The SD card is located on the CPU board. Carefully slide the SD card into or out of the holder. Only FAT-16 or FAT-32 format SD cards can be used: SDHC cards cannot be used.

Note Make sure that the status of the SD card is set to *Remove* (see chapter 6.4.1).

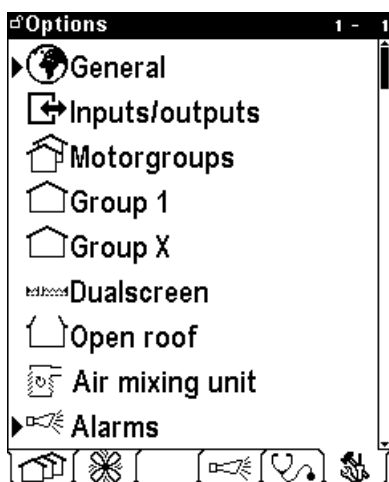


7 Installation procedure

Once the Orion-MC has been connected in accordance with the connection diagrams (see chapter 9) then you can enable the required options. Pressing the **F6** key when the main group screen is displayed opens the following screen:



This screen can be used to view and/or change the options and switch the Orion-MC on or off with the software. Moving the cursor to *View options* or *Change options* and then pressing **▶** opens the option groups, provided that these are not secured with an access code. You always have access to *View options*, even when changes cannot be made without an access code. The access code is set in the *General* group (see chapter 7.1). The option groups that can be configured during the installation procedure are:



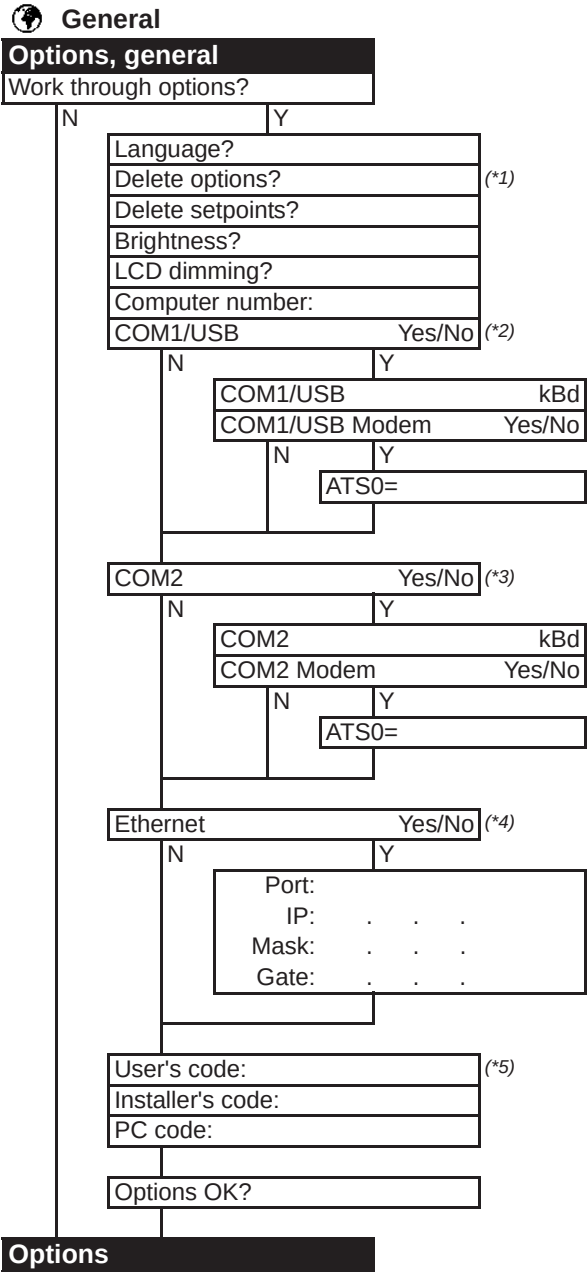
Note Your screen may differ from the screen shown above: solely the option groups of the available modules are displayed!

You can navigate through the various option groups with **▼** or **▲** and then **▶**. In the options screen you can use **▼** and **▲** to navigate through all questions: press **▶** to change the option settings.

Fill in the configuration of each option group in the charts shown on the following pages.

7.1 General options

The general options include the language, communication settings and access codes.



- (*1) Set the required language. If so required, you can delete existing options and setpoints. You can adjust the brightness of the screen. Activating LDC dimming will automatically dim the screen after 1 minute of no activity.
- (*2) Enable the COM1 or USB port here. In addition, set the communication speed (default: 38.4 kBd). Under Answer (ATS0=), set the number of rings before the modem answers.
- (*3) Enable the COM2 port here. In addition, set the communication speed (default: 38.4 kBd). Under Answer (ATS0=), set the number of rings before the modem answers.
- (*4) Enter the network settings here.
 Port = the port number.
 IP = the IP address of the Orion
 Mask = Definition of the local network.
 Gate = IP address of the router.
 Use the ◀ and ▶ keys to browse through the settings. You can use the ▼ and ▲ keys to change the figures.
- (*5) A password consists of digits. When the password is to be entered the keys of the Orion are assigned the numerical values displayed on the screen (F1=1, F2=2, ▲=7, etc.). Confirm the entry with **OK**. Under Repeat, enter your selected password again.
 Passwords may not begin with 0 and must consist of at least 1 and at most 4 digits.
 Entering 0 (▼) disables the password function.
 The user's access code can be used to disable access to the control settings and installation procedure. The settings are automatically disabled once you have not pressed any key for 10 minutes.
 The installer's access code can be used to secure access to the installation procedure.
 The PC code secures logging in via PC software.

7.2 IO module settings

You use the IO modules options to configure the inputs and outputs.

Example You have connected an analog control signal from the main controller to a IO-07-07 module. You now need to specify the signal received by this input in the IO options.



Options: IO

IO-07-07[1] (*6)

IO-08-09[-]

IO-EKU[-]

IO-H1MC[-]

IO-H1MC[-]

(*6) The standard Orion-MC is supplied with one integral IO-07-07. You can connect additional IO modules via CAN Local (see chapter 9.3.1).

Options: IO-07-07

Work through options?

N Y

Number of IO module: 1..16

Analog inputs: (*7)

Analog outputs: (*8)

Digital outputs: (*9)

RS485 Yes/No (*10)

N Y

Baud rate: kBd

Parity: Non / Even / Odd

Alarm contact Yes/No (*11)

Copy options to IO number: (*12)

Options OK?

Y

Options: IO

(*7) Enter the CAN Local address configured on the IO module here.

(*8) Press **OK** to set the minimum and maximum of the output (voltage).

(*9) Press **OK** to test the output with the **▼▲** keys.

(*10) Enable the RS485 communication (ebmBus or Modbus) of the IO module here. Set the data speed and parity to the correct values.

- ebmBus; 9.6kBd, no parity
- Modbus Ziehl-Abegg; 19.2kBd, even parity
- Modbus EBM; 19.2kBd, even parity

(*11) Yes = the alarm contact of this module is de-energised in the event of a hard alarm. No = the alarm contact is de-energised only when the CAN-Local communication is broken or in the event of a power failure.

(*12) The options of identical IO modules are often the same. You can use *Copy options* to copy the IO module's settings.

Enter the CAN Local address of the new IO module and confirm with **OK**. Confirm again with **OK**. The new IO module will then be created with the same options. You can then continue immediately with the options of the newly created IO module.

Options: H1MC/H2MC/EKU (*13)

Work through options?

N

Y

Number of IO module: (*14)

-

1..32

Motor control:

(*15)

Alarm contact

Yes/No

(*16)

Options OK?

Y

Options: IO

(*13) The options for the EKU, H2MC and H1MC are virtually identical.

(*14) Enter the CAN Local address configured on the IO module here.

(*15) Enter the correct type of motor.
Window motor (100% = open, 0% = closed)
Screen motor (100% = open, 0% = closed)
You can also view the current setting for the position here.

(*16) Yes = the alarm contact of this module is de-energised in the event of a hard alarm. No = the alarm contact is de-energised only when the CAN-Local communication is broken or in the event of a power failure.

7.3 Options for air mixing unit

Options for the control of air mixing units.

Options: IO-07-07

Work through options?

N

Y

Number of groups: (*17)

Inner/outer flap

|continue from B

Recirculation flap

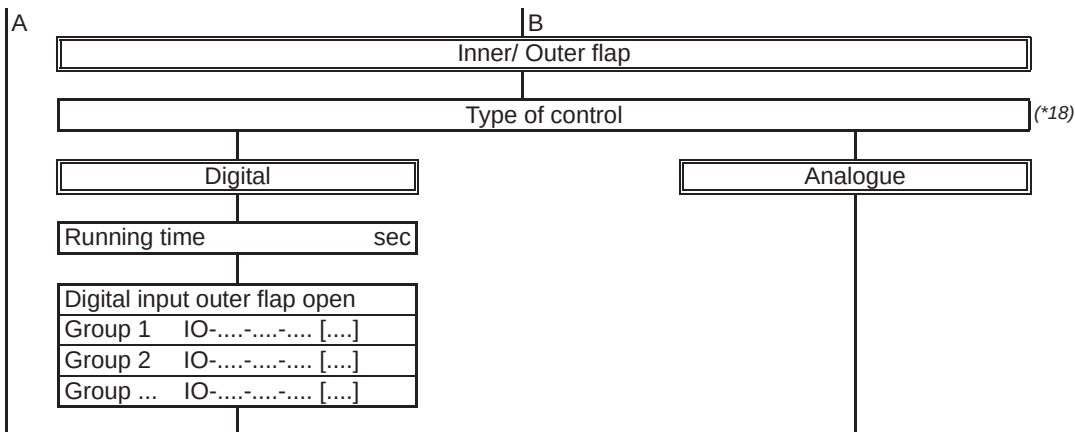
|continue from C

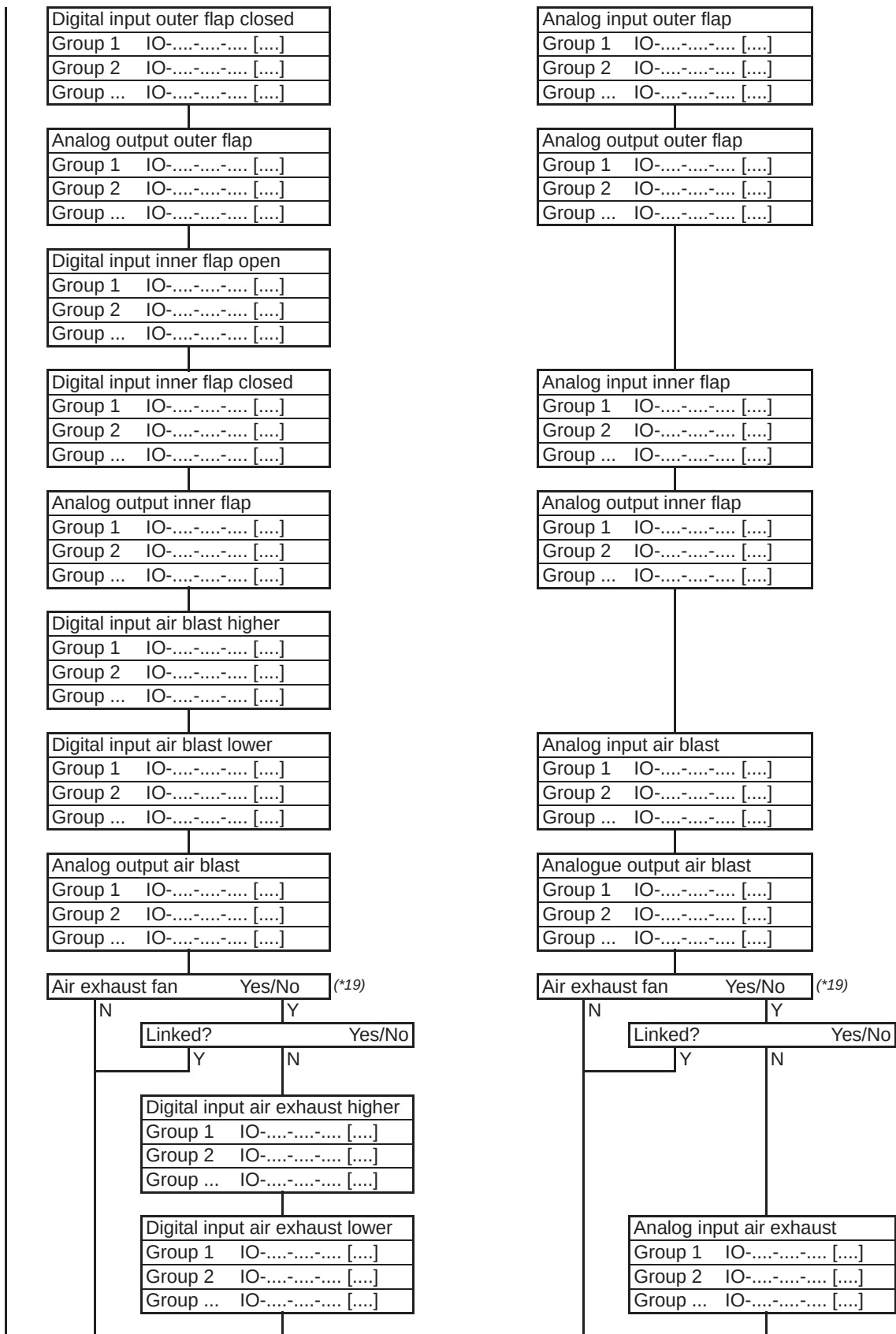
(*17) Configure the number of groups controlled by the main controller. Select the type of air mixing units that must be controlled (see also Appendix 2.2).

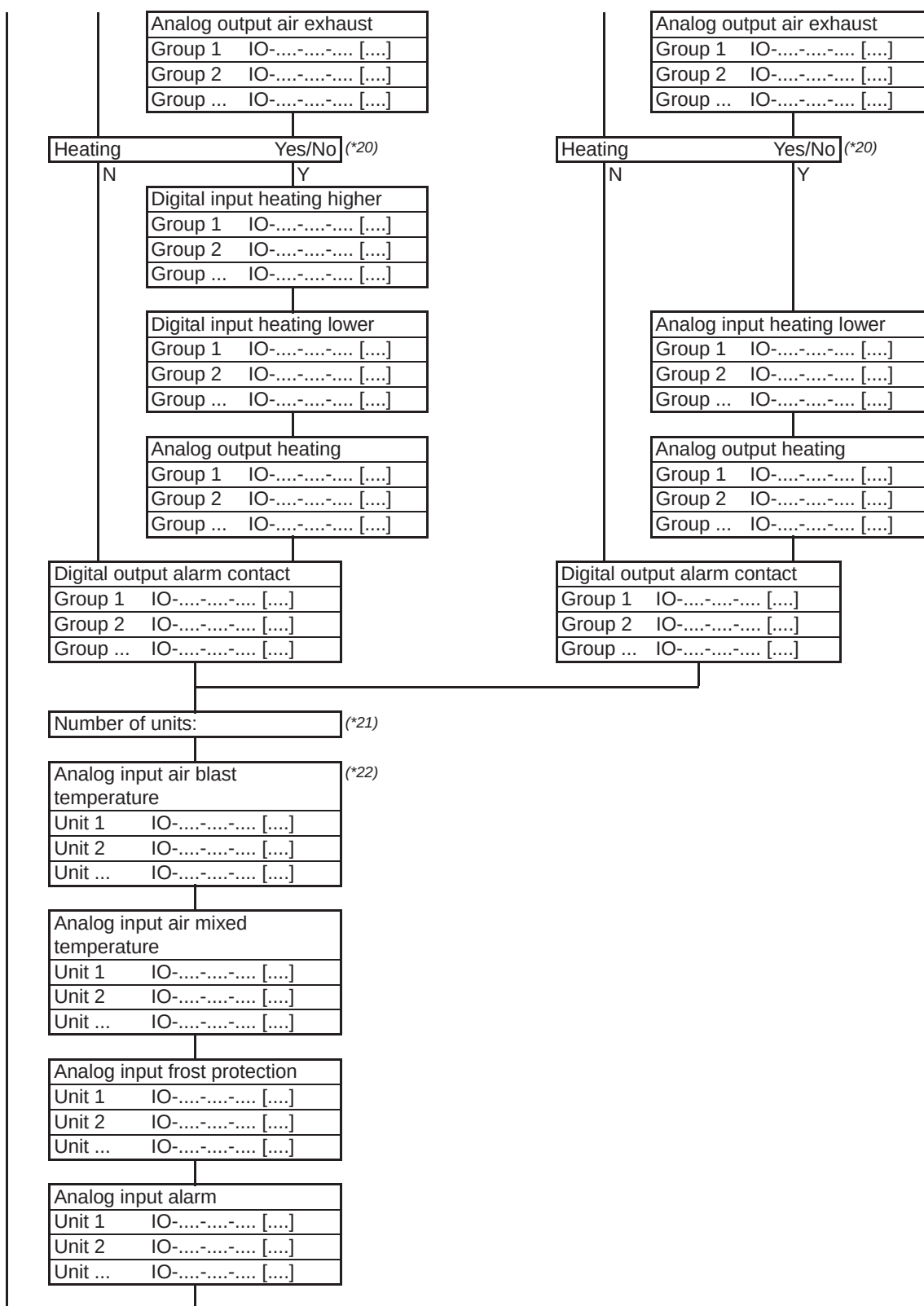
Inner/outer flap: The inner and outer flaps are controlled separately, and an upper flap is not possible.

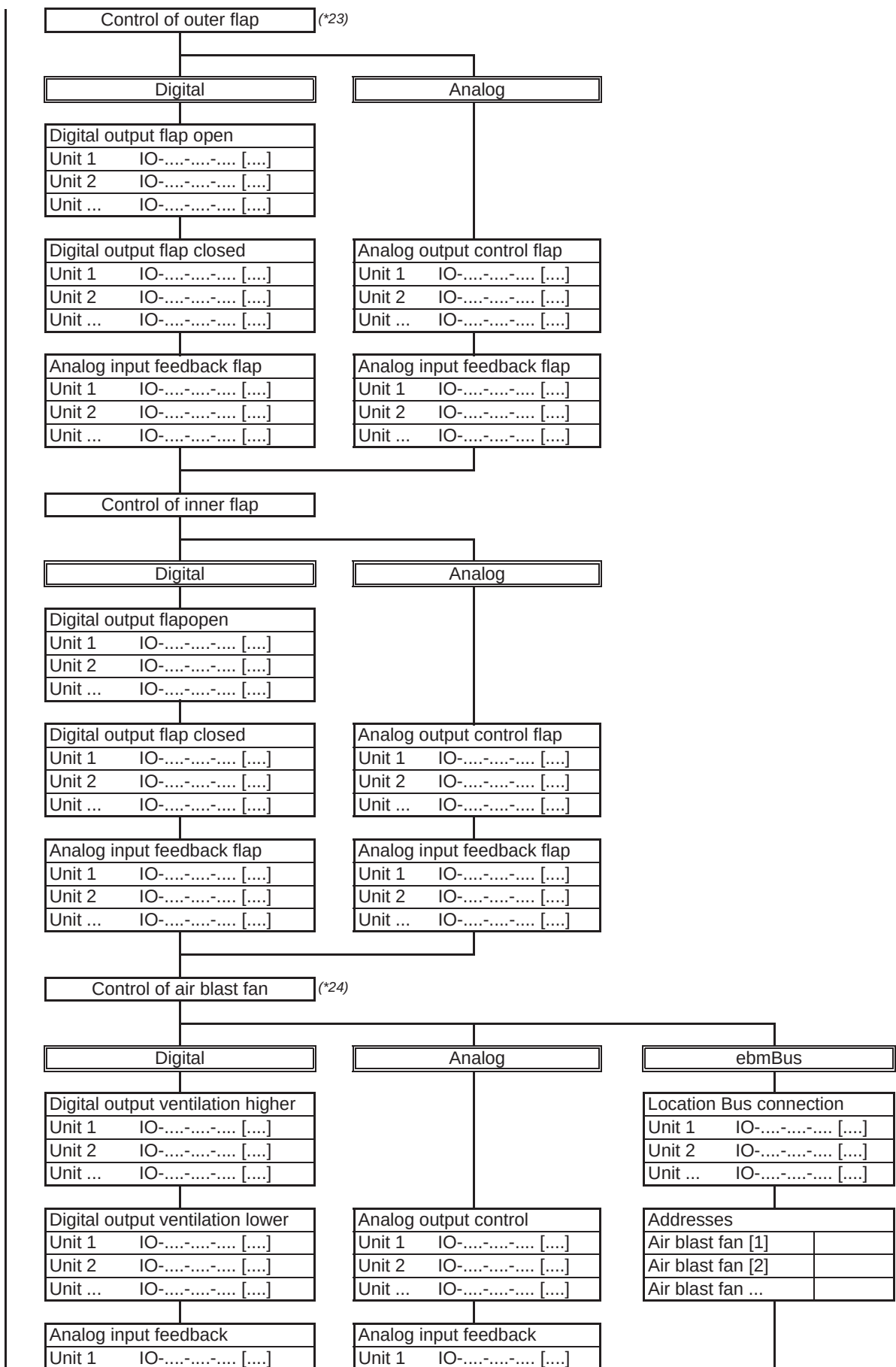
Recirculation flap: The inner and outer flaps are linked and controlled in one control loop. As a result, you can have an upper flap controlled (option).

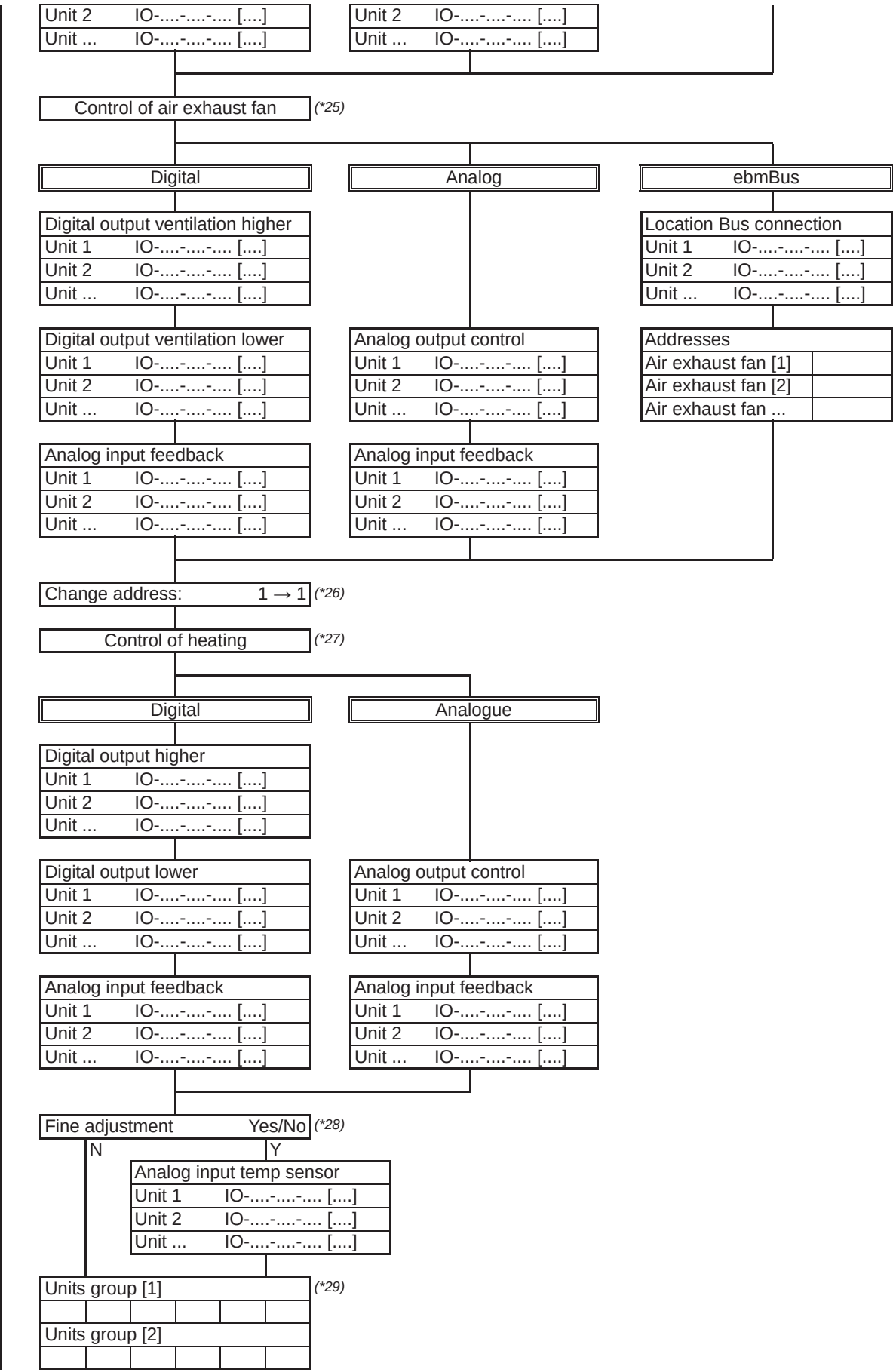
Since the installation procedure for the two types differs in some respects the explanation of the installation procedure is divided into *Inner/outer flap (B)* and *Recirculation flap (C)*.









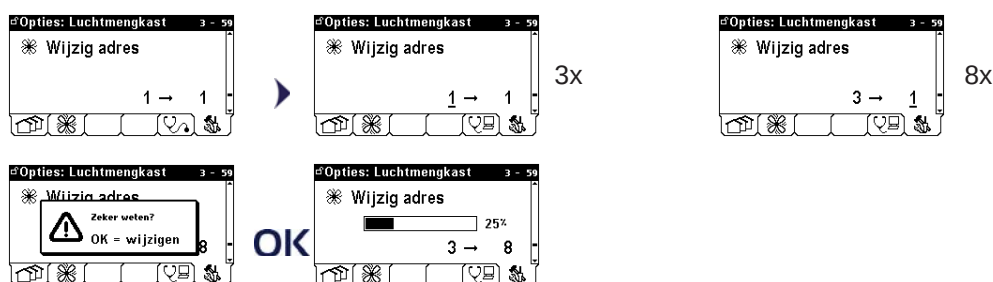


Units group [...]					
Options OK?					
Y					

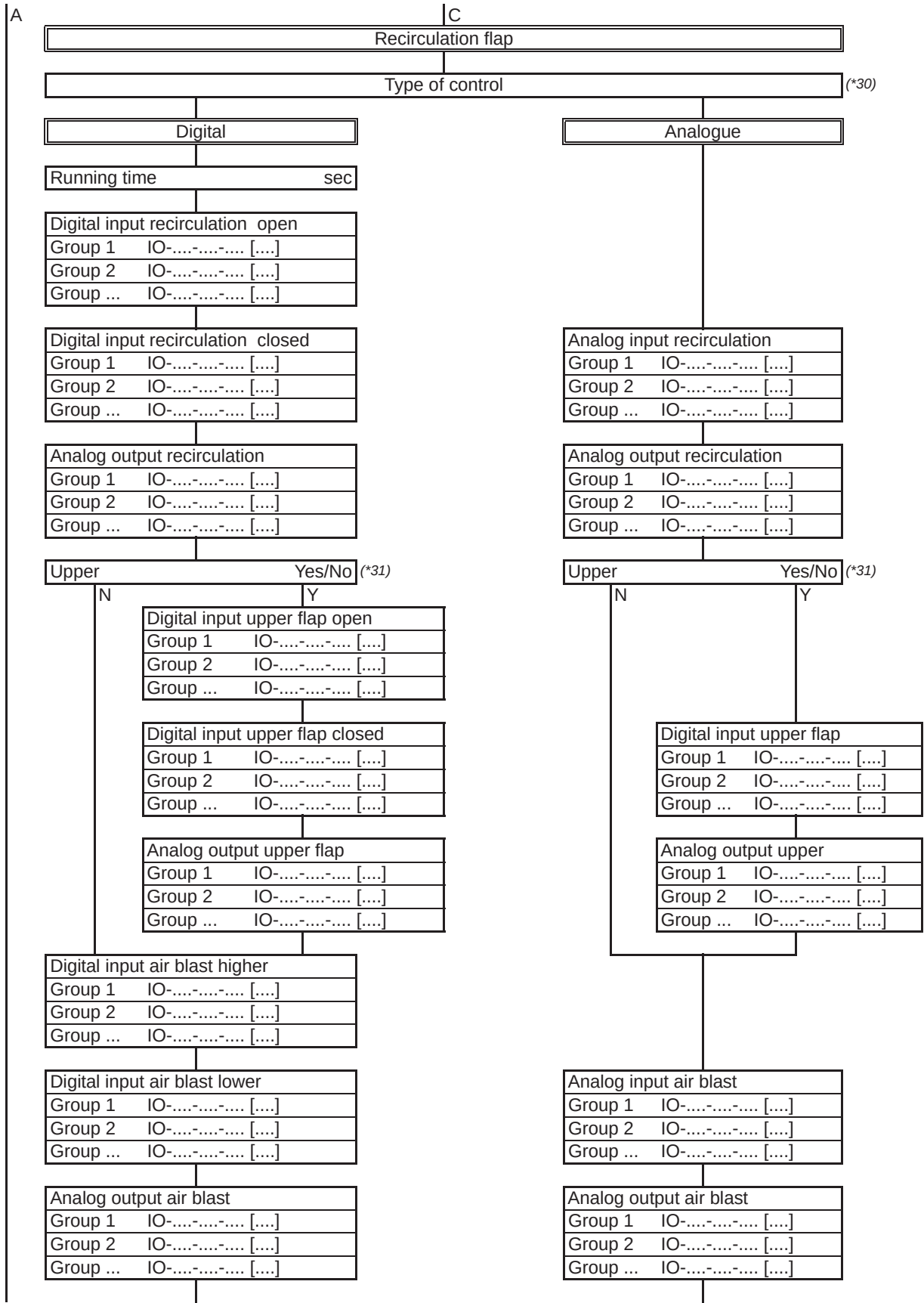
Options

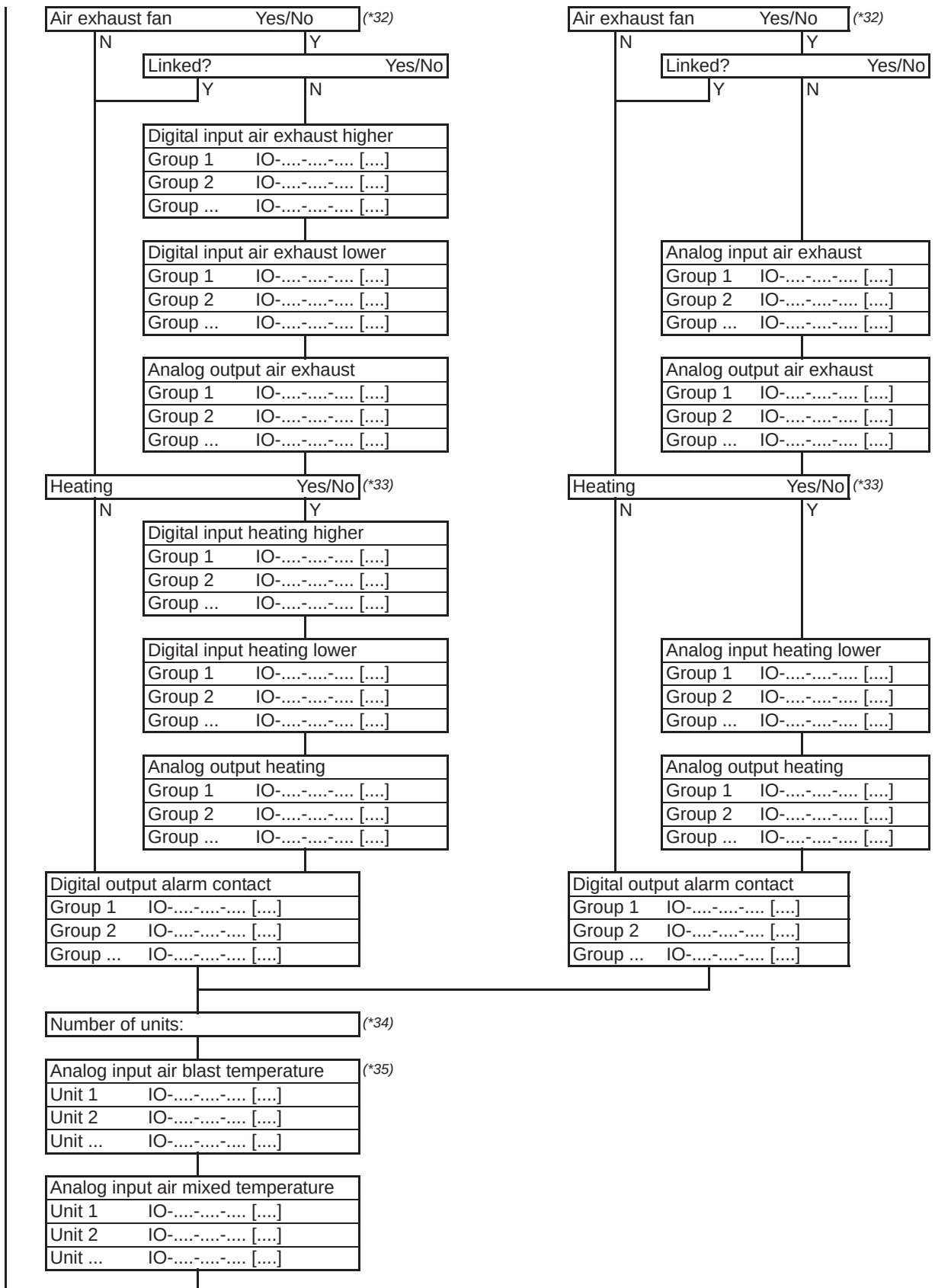
Note The options under (*18) (*19) and (*20) are not applicable to CANopen or BACnet communication.

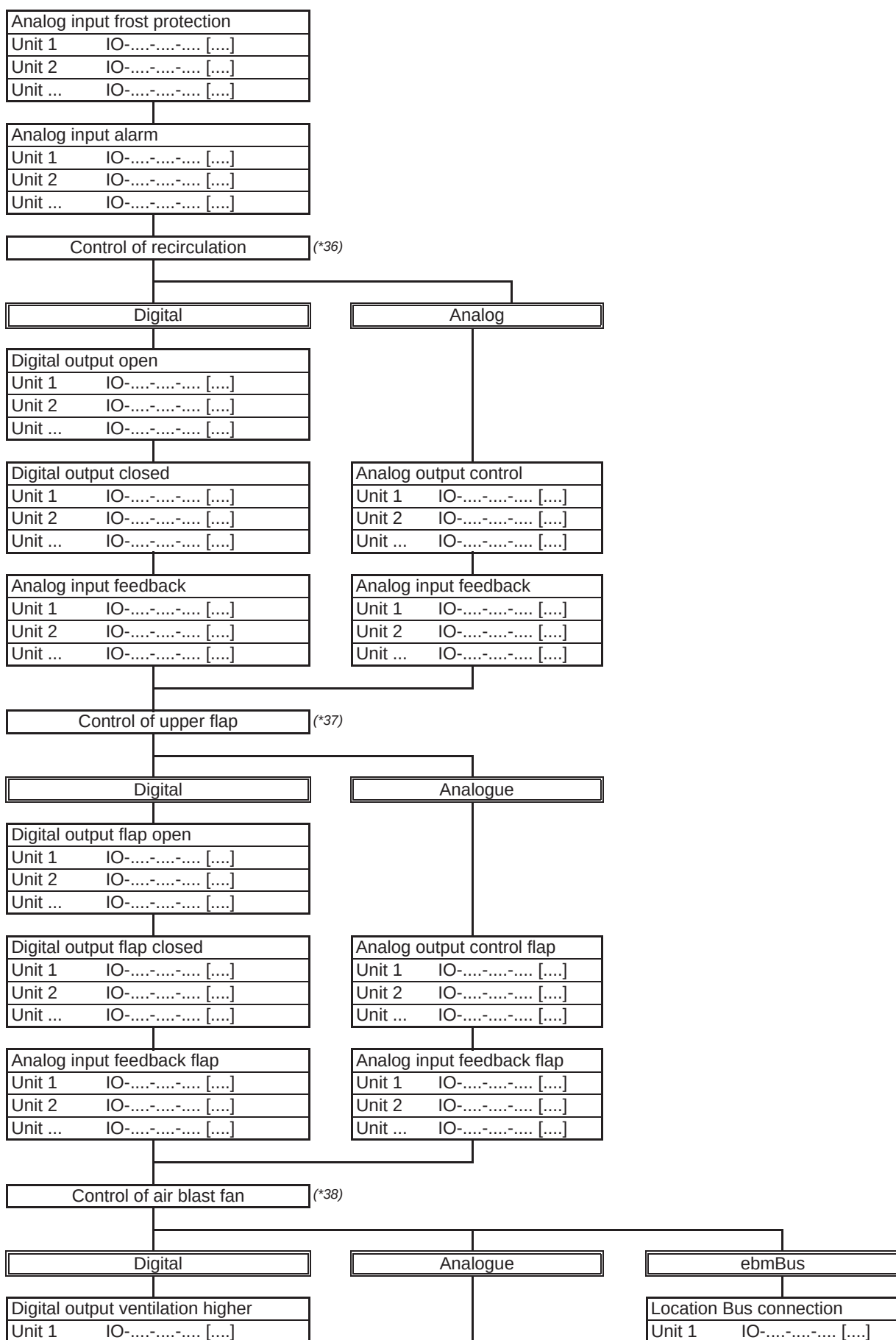
- (*18) Configure the signals to and from the **main controller** for the outer flap and inner flap of each group of air mixing units. The main controller transmits *digital* or *analogue* signals for the control of the flap. In addition, the Orion-MC transmits an analogue signal (= virtual position) to the main controller. Configure the signals to and from the **main controller** for the air blast fan of each group of air mixing units. The main controller transmits *digital* or *analogue* signals for the control of the fan. In addition, the Orion-MC transmits an analogue signal (= virtual fan position) to the main controller.
- (*19) Option: Configure the signals to and from the **main controller** for the air exhaust fan of each group of air mixing units.
- (*20) Option: Configure the signals to and from the **main controller** for the heating of each group of air mixing units.
- (*21) Enter the total number of air mixing units used (= units).
- (*22) Option: Enter the inputs to which the temperature sensors are connected here.
- (*23) Configure the signals to and from the **air mixing unit** for the inner flap and outer flap of each group of air mixing units.
The Orion-MC transmits *digital* or *analogue* signals for the control of the flap. In addition, the Orion-MC transmits an analogue feedback signal (= actual position) to the Orion-MC.
- (*24) Configure the signals to and from the **air mixing unit** for the air blast fan of each group of air mixing units. The fans can be controlled by a *digital* or *analogue* signal or via an *ebmBus*. Enter under the *ebmBus* the code of the IO module to which the fans are connected and enter the number of the fan (= address) for the relevant air mixing unit. In addition, with digital or analogue control the fan transmits an analogue feedback signal (= actual fan position) to the Orion-MC.
- (*25) This option is displayed only when the option under (*19) is enabled. Configure the signals to and from the **air mixing unit** for the air exhaust fan of each group.
- (*26) This option is displayed only when *ebmBus* is entered under (*24) or (*25). Configure the addresses of the fans here (Each fan must be assigned a unique address.) Example: changing address 3 to address 8.

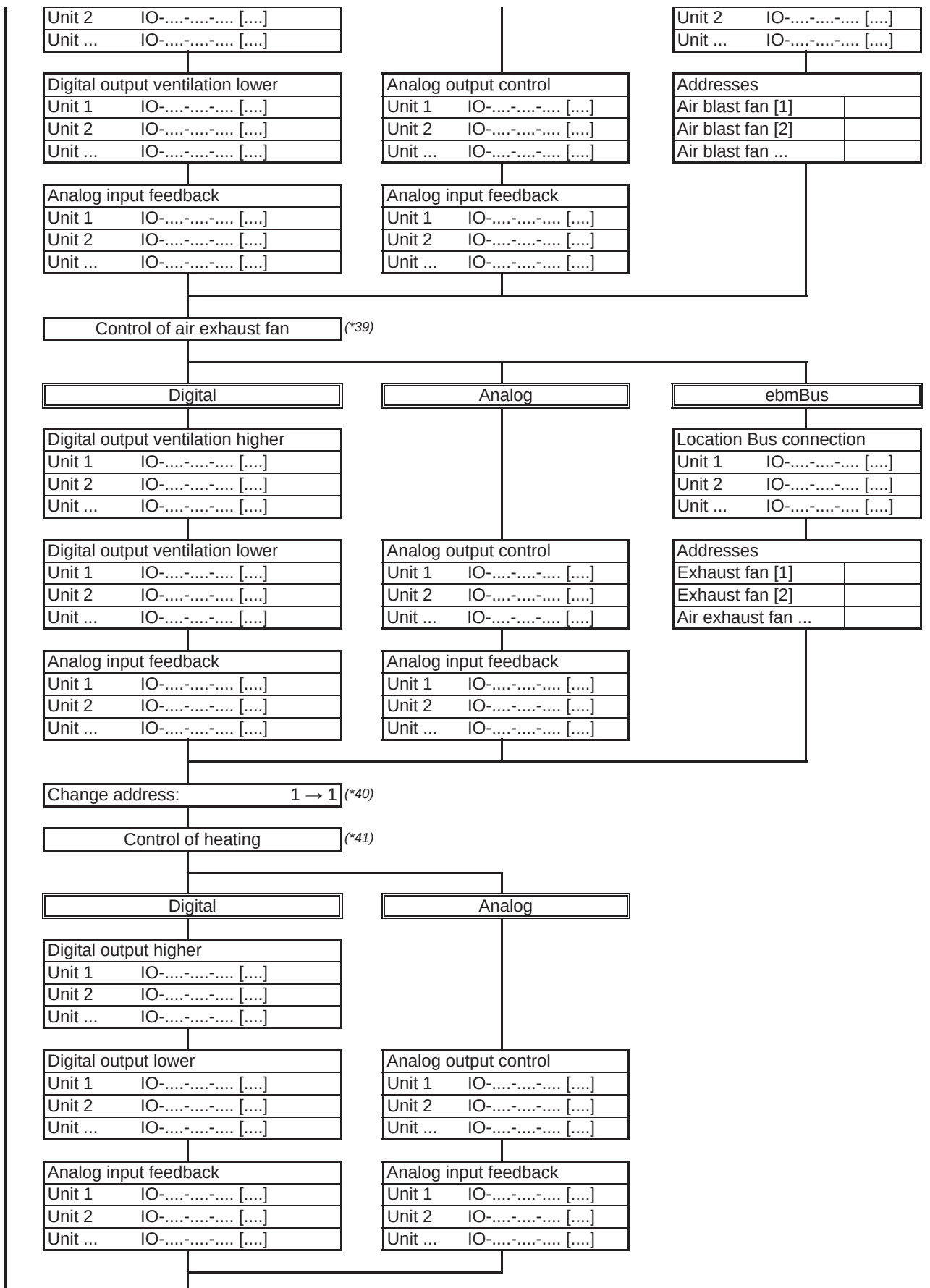


- (*27) This option is displayed only when the option under (*20) is enabled. Configure the signals to and from the **air mixing unit** for the heating here.
- (*28) The fine adjustment option is available only when the air mixing units are equipped with heating and a temperature sensor. The main controller will then correct the heating of individual air mixing units on the basis of the *air blast temperature* of one air mixing unit (the master).
Without *fine adjustment* the heating of the other air mixing units (slaves) will be controlled to the same position. When the *fine adjustment* option is enabled the Orion-MC will adjust the air blast temperature of the slave units if so required. When the *air blast temperature* of the slave is lower than the *air blast temperature* of the master then the slave's heating will be increased. When the *air blast temperature* of the slave is higher than the *air blast temperature* of the master then the slave's heating will be decreased. See also Annex 2.5.
- (*29) Division of the units between the groups. Uncheck the ☒ next to the unit numbers that are not part of the relevant group.









Fine adjustment		Yes/No	(*42)
N		Y	
Analog input temp sensor			
Unit 1	IO-....-....-....	[....]	
Unit 2	IO-....-....-....	[....]	
Unit ...	IO-....-....-....	[....]	
Units group [1] (*43)			
Units group [2]			
Units group [...]			
Options OK?			
		Y	

Options

Note The options under (*30) (*32) and (*33) are not applicable to CANopen or BACnet communication.

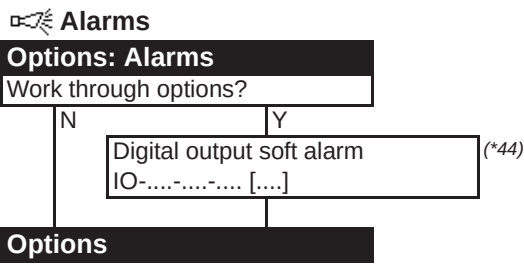
- (*30) Configure the signals to and from the **main controller** for the recirculation flap of each group of air mixing units. The main controller transmits *digital* or *analogue* signals for the control of the flaps. In addition, the Orion-MC transmits an analogue signal (= virtual flap position) to the main controller.
- Configure the signals to and from the **main controller** for the air blast fan of each group of air mixing units. The main controller transmits *digital* or *analogue* signals for the control of the fan. In addition, the Orion-MC transmits an analogue signal (= virtual fan position) to the main controller.
- (*31) Option: Configure the signals to and from the **main controller** for the upper flap of each group of air mixing units.
- (*32) Option: Configure the signals to and from the **main controller** for the air exhaust fan of each group of air mixing units.
- (*33) Option: Configure the signals to and from the **main controller** for the heating of each group of air mixing units.
- (*34) Enter the total number of air mixing units used (= units).
- (*35) Option: Enter the inputs to which the temperature sensors are connected here.
- (*36) Configure the signals to and from the **air mixing unit** for the recirculation flap of each group of air mixing units. The Orion-MC transmits *digital* or *analogue* signals for the control of the . In addition, the flap transmits an analogue feedback signal (= actual position) to the Orion-MC.
- (*37) This option is displayed only when the option under (*31) is enabled. Configure the signals to and from the **air mixing unit** for the upper flap of each group.
- (*38) Configure the signals to and from the **air mixing unit** for the air blast fan of each group of air mixing units. The fans can be controlled by a *digital* or *analogue* signal or via an *ebmBus*. Enter under the *ebmBus* the code of the IO module to which the fans are connected and enter the number of the fan (= address) for the relevant air mixing unit. In addition, with digital or analogue control the fan transmits an analogue feedback signal (= actual fan position) to the Orion-MC.
- (*39) This option is displayed only when the option under (*32) is enabled. Configure the signals to and from the **air mixing unit** for the air blast fan of each group.
- (*40) This option is displayed only when *ebmBus* is entered under (*38) or (*39). Configure the addresses of the fans here (Each fan must be assigned a unique address.) Example: changing address 3 to address 8.



- (*41) This option is displayed only when the option under (*33) is enabled. Configure the signals to and from the **air mixing unit** for the heating here.

- (*42) The fine adjustment option is available only when the air mixing units are equipped with heating and a temperature sensor. The main controller will then correct the heating of individual air mixing units on the basis of the *air blast temperature* of one air mixing unit (the master). Note! You can choose from all the available temperature sensors, select the correct air blast temperature sensor.
Without *fine adjustment* the heating of the other air mixing units (slaves) will be controlled to the same position. When the *fine adjustment* option is enabled the Orion-MC will adjust the air blast temperature of the slave units if so required. When the *air blast temperature* of the slave is lower than the *air blast temperature* of the master then the slave's heating will be increased. When the *air blast temperature* of the slave is higher than the *air blast temperature* of the master then the slave's heating will be decreased. See also Appendix 2.2.
- (*43) Division of the units between the groups. Uncheck the ☒ next to the unit numbers that are not part of the relevant group.

7.4 Alarms



- (*44) You can assign a contact to a soft alarm here. This contact will then be de-energised in the event of a soft alarm.

8 Technical specifications

8.1 Orion-UPM-30

IMPORTANT!

- When an installation has two or more components that are fed separately, check if all components can operate at the same mains voltage.
- Always replace a defective fuse with a fuse of the same brand, type and fuse rating.

Electrical

Types of mains supply	: Fixed wiring
Operating area	: Indoor, normal, dry (not suitable for wet areas)
Operating conditions	: Continuous operation
Mains power supply	: 100-240 VAC
Frequency	: 50-60 Hz
Power consumption	: 0.75-0.5 A
PCB fuse	: T800mA (5 mm x 20 mm)
Pre-fuse	: Max. 16 A
Insulation	: Class 1 (IEC 61140)
Transient over-voltage	: Category II (IEC 60664-1)
Pollution degree	: Category 2 (IEC 61010-1)

Mechanical

Mobility	: Fixed position (the product is mounted to the wall)
Operating temperature range	: 0 - 40 °C
Relative humidity	: Max. 80%
Installation height	: Max. 2000 m above sea level
Dimensions (H x W x D)	: 220 x 165 x 105 mm
Housing	: IP54, ABS
Weight	: ± 1 kg

Power supply (UPM-30)

Voltage 24 VDC $\pm 2.5\%$	: Max. 800 mA
----------------------------	---------------

Communication

CAN Local	: Max. 500 m @ 100 kbps
CAN Backbone	: Max. 500 m @ 125 kbps
RS232	: Max. 15 m @ 38400 baud
USB	: Max. 5 m @ 115 kbps
Ethernet	: Max. 100 m @ 10 Mbps

Cables

CAN Local / CAN Backbone	: Shielded twisted pair, 1x2x0.64 mm ² (CAN-BK cable)
RS232	: MSK-9
USB	: USB with USB-B mini plug and USB-A plug

Ethernet : UTP cat 5e

Compliance

EU Directives : See the Declaration of Conformity for this product

8.2 CAN-IO-07-07

IMPORTANT!

- This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.
- When an installation has two or more components that are fed separately, check if all components can operate at the same mains voltage.
- Always replace a defective fuse with a fuse of the same brand, type and fuse rating.
- Always prevent ground loops when you install the product. If possible, provide galvanic isolation.

Electrical

Types of mains supply	: Fixed wiring
Operating area	: Indoor, normal, dry (not suitable for wet areas)
Operating conditions	: Continuous operation
Mains power supply	: 230 VAC $\pm 10\%$
Frequency	: 50/60 Hz
Power consumption	: max. 15 VA
PCB fuse	: T100mA (5 mm x 20 mm)
Insulation	: Class 1 (IEC 61140)
Transient over-voltage	: Category II (IEC 60664-1)
Pollution degree	: Category 2 (IEC 61010-1)

Mechanical

Mobility	: Fixed position (the product is mounted to the wall)
Operating temperature range	: 0 - 40 °C
Relative humidity	: Max. 80%
Installation height	: Max. 2000 m above sea level
Dimensions (H x W x D)	: 220 x 165 x 105 mm
Housing	: IP54, ABS
Weight	: ± 1 kg

Analog inputs (4)

Temperature sensor (PT1000)	: -40 - 100 °C ± 0.5 °C
Potentiometer	: 5 k Ω
0-20 mA	: $R_i = 250 \Omega$
0-10 V	: $R_i = 50 \text{ k}\Omega$

Digital inputs (3)

Contact / counter	: NPN / PNP sensor, 24 VAC/DC, 8 mA, max. 10 Hz
-------------------	---

Analog outputs (2)

0-10 VDC	: 0-10 V, max. 1 mA
----------	---------------------

Digital outputs (4 + 1 alarm output)

Relay output K1-K4	: 0.5 A, 230 VAC
Alarm output	: 0.5 A, 24 VAC/DC

Power supply, 24 VDC

Voltage 24 VDC -30% +10%	: Max. 300 mA
--------------------------	---------------

Communication

CAN Local	: Max. 500 m @ 100 kbps
RS485	: Max. 1200 m (depends on the speed)

Cables

CAN Local	: Shielded twisted pair, 1x2x0.64 mm ² (CAN-BK cable)
RS485	: Shielded twisted pair, 1x2x0.64 mm ² (RS485 cable)

Compliance

EU Directives	See the Declaration of Conformity for this product
---------------	--

9 Connection diagrams

9.1 Connection sequence

Read these instructions thoroughly before you begin work on the assembly and connections.

This product should be installed by an approved installer with certification according to the relevant prevailing standards (for example, ANSI, BS, ISO). The installer must possess the knowledge of and experience with electronic control equipment required to install the Orion-MC.

IMPORTANT! For your safety, you must work through the installation procedure in the following sequence.

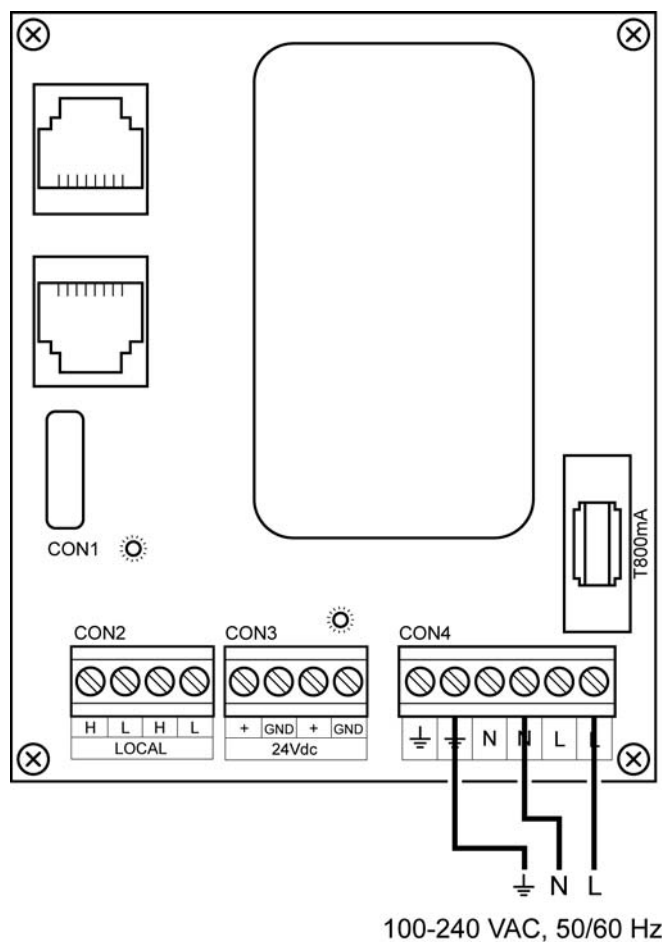
1. Install the necessary equipment (see also Appendix 2).
2. Install the necessary IO modules (see also Appendix 2).
3. Read in all motors and equipment with position feedback: these are the motors controlled using the IO modules such as the H1MC, H2MC and CAN-IO-EKU modules. Reading in these motors and equipment will ensure that they do not go out of range during the installation of the Orion-MC.
4. If possible, set all IO modules and connected equipment in manual operation mode: this will make sure that the motors and equipment to be controlled by the unit will not be started unintentionally during the installation of the Orion-MC.

Note The following connection diagrams explain the communication with the main controller, and do not extend to the connection of options (such as temperature sensors, RS485 communication, etc.). Information about the connection of options is enclosed in the relevant CAN IO module manual. In addition, see the examples of connections in Appendix 2.

9.2 Orion-UPM-30

IMPORTANT!

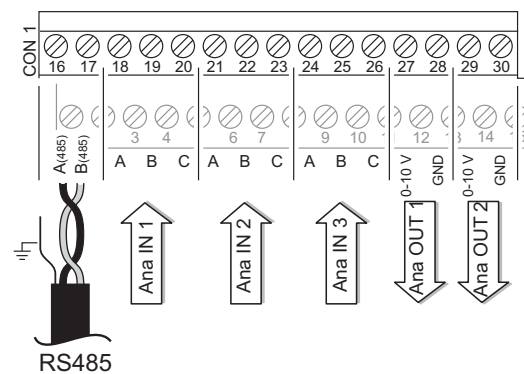
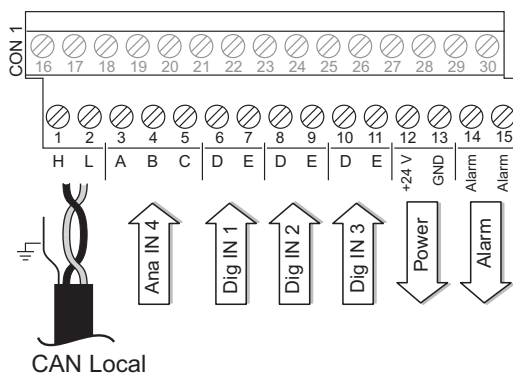
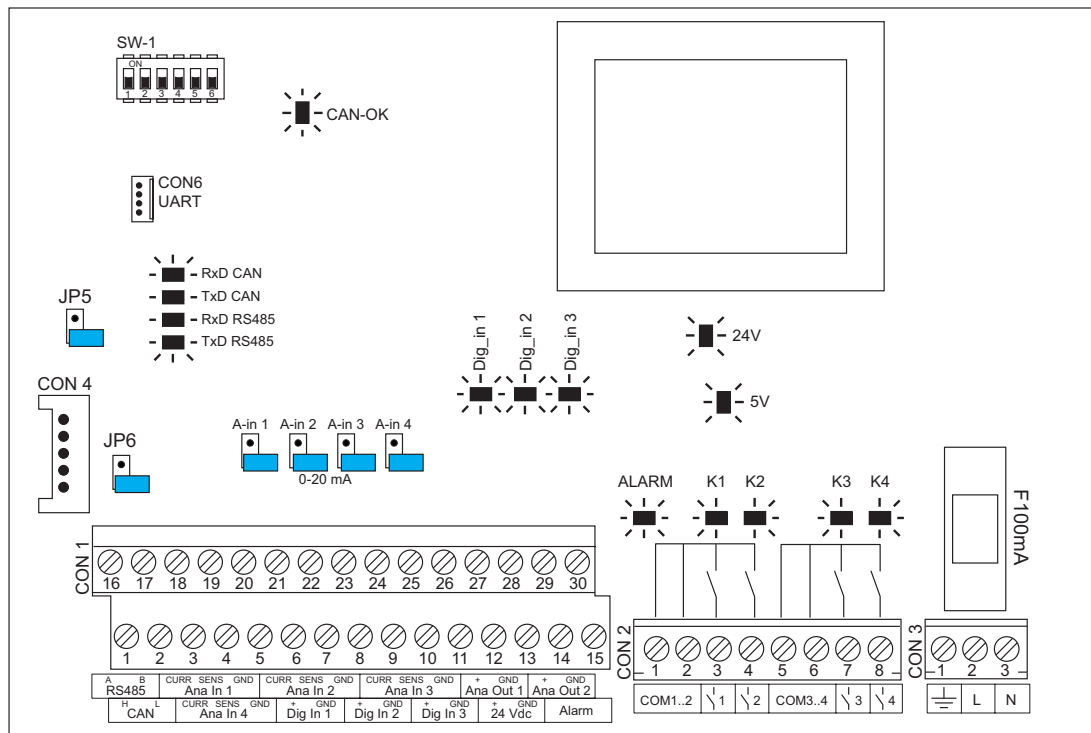
- Always disconnect the power supply to the product before you do work on electrical parts inside the product. When this is not possible, be very careful that you do not touch live parts.
- This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.
- When an installation has two or more components that are fed separately, do a check if all the components can operate at the same mains voltage.
- All the electrical parts inside this product are susceptible to electrostatic discharge (ESD).
 - Always disconnect the power supply to the product before you do work on electrical parts inside the product.
 - Always wear an ESD wrist strap that is connected to the main earthing system. See IEC61340-5-2 for more information on precautions with regard to ESD.
- Always prevent ground loops when you install the product. If possible, provide galvanic isolation.
- A mix of voltages at the relay contacts can damage your product.
 - Connect the digital outputs (relay contacts) to the same electrical potential.
 - Do not mix mains and low-voltage signals.



9.3 CAN-IO-07-07P

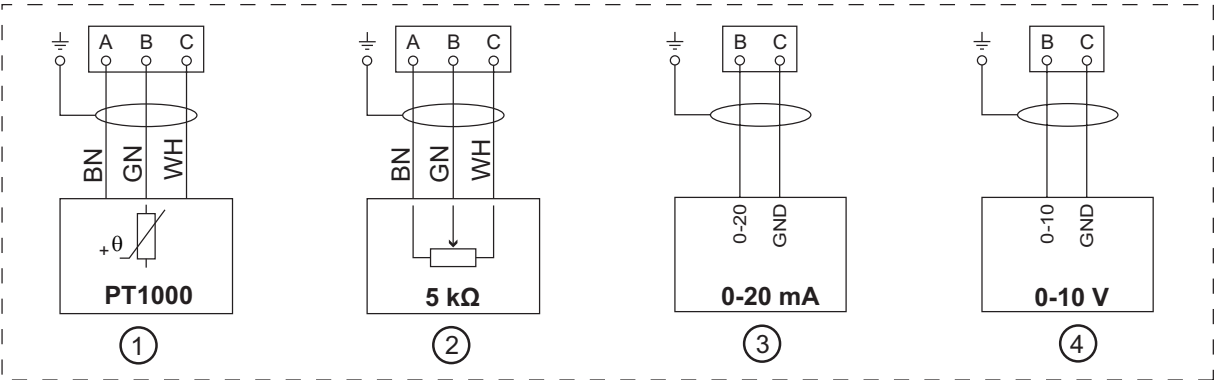
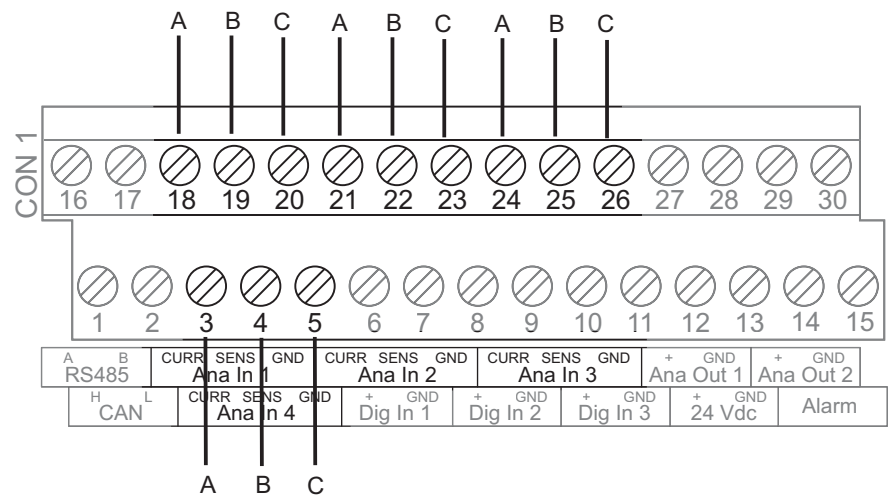
IMPORTANT!

- Always disconnect the power supply to the product before you do work on electrical parts inside the product. When this is not possible, be very careful that you do not touch live parts.
- This product can have more than one power source. Always disconnect all the power sources before you do work on live or exposed parts of the product.
- When an installation has two or more components that are fed separately, do a check if all the components can operate at the same mains voltage.
- All the electrical parts inside this product are susceptible to electrostatic discharge (ESD).
 - Always disconnect the power supply to the product before you do work on electrical parts inside the product.
 - Always wear an ESD wrist strap that is connected to the main earthing system. See IEC61340-5-2 for more information on precautions with regard to ESD.
- Always prevent ground loops when you install the product. If possible, provide galvanic isolation.
- A mix of voltages at the relay contacts can damage your product.
 - Connect the digital outputs (relay contacts) to the same electrical potential.
 - Do not mix mains and low-voltage signals.



9.3.1 Analog inputs

The analog inputs are intended for the control signals from the main controller. See Appendix 2.1 for examples of connections. The analog inputs are assigned during the installation procedure.



Legend:

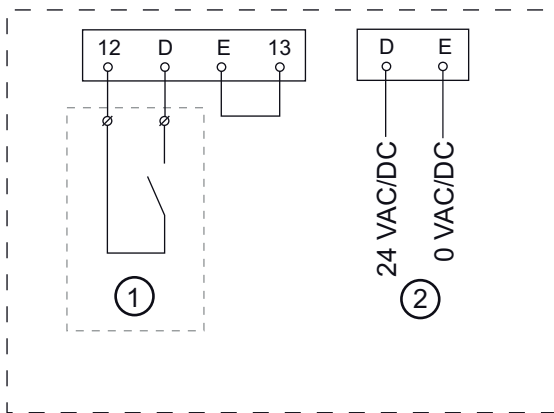
1. Temperature
2. Potentiometer
3. Analog input 0-20 mA
4. Analog input 0-10 V

Note Make sure that jumper A-in (x) for the 0-20 mA analog input is set correctly.

Note the inputs in use in the following table.

Analog input	Terminal	Selection								Description
		Window	Screen	Ventilation	Flap	Heating	Alarm	Frost guard	Temperature	
1	19 - 20									
2	22 - 23									
3	25 - 26									
4	4 - 5									

9.3.2 Digital inputs

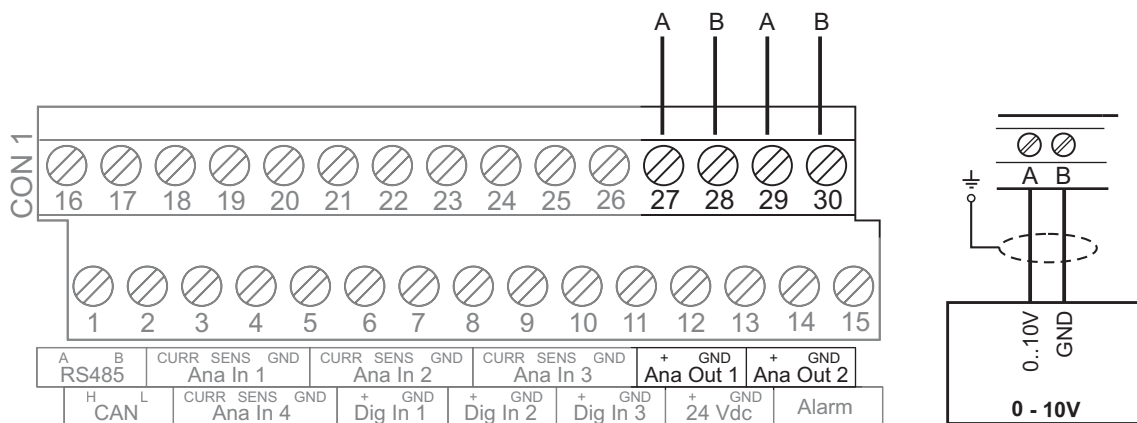


Legend:

1. External contact, for example a frost guard or motor protection
2. External control signals

9.3.3 Analog outputs

The analog outputs are intended for the feedback to the main controller. See Appendix 2.1 for examples of connections. The analog outputs are assigned during the installation procedure.

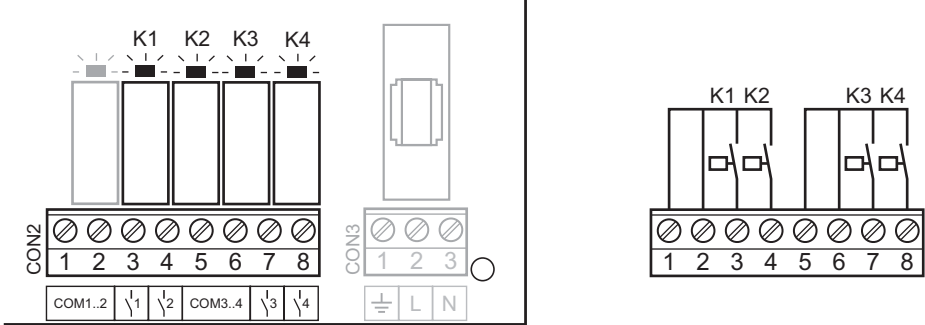


Note the outputs in use in the following table. In addition, state the range of the output (e.g., 1-9V).

Analog output	Terminal	Selection:							Range	Description
		Window position	Screen position	Fan speed	Flap position	Heater intensity				
1	27 - 28								Min / Max	
2	29 - 30									

9.3.4 Digital outputs

The digital outputs are intended for the feedback to the main controller. See Appendix 2.1 for examples of connections. The digital outputs are assigned during the installation procedure.

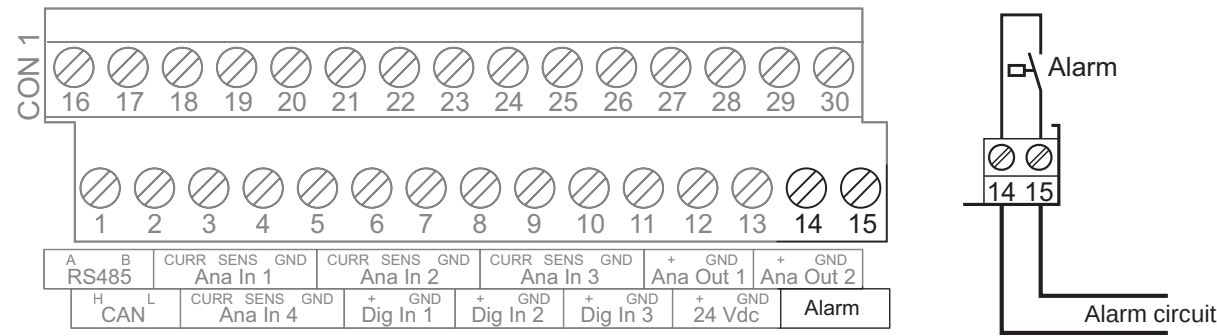


Note the outputs in use in the following table.

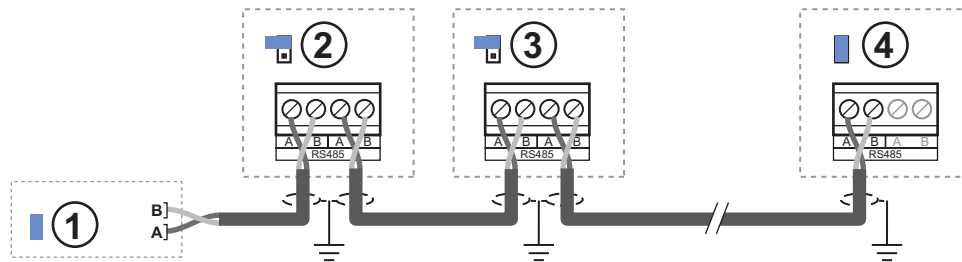
Digital output	Terminal	Selection							Description
		Alarm	Ventilation	Flap	Heating				
1	1/2 - 3								
2	1/2 - 4								
3	5/6 - 7								
4	5/6 - 8								

9.3.5 Alarm contact

The no-voltage alarm contact is interrupted in the event of an alarm or a power failure. You can include the alarm contact in an alarm circuit.



9.3.6 RS485



Legend:

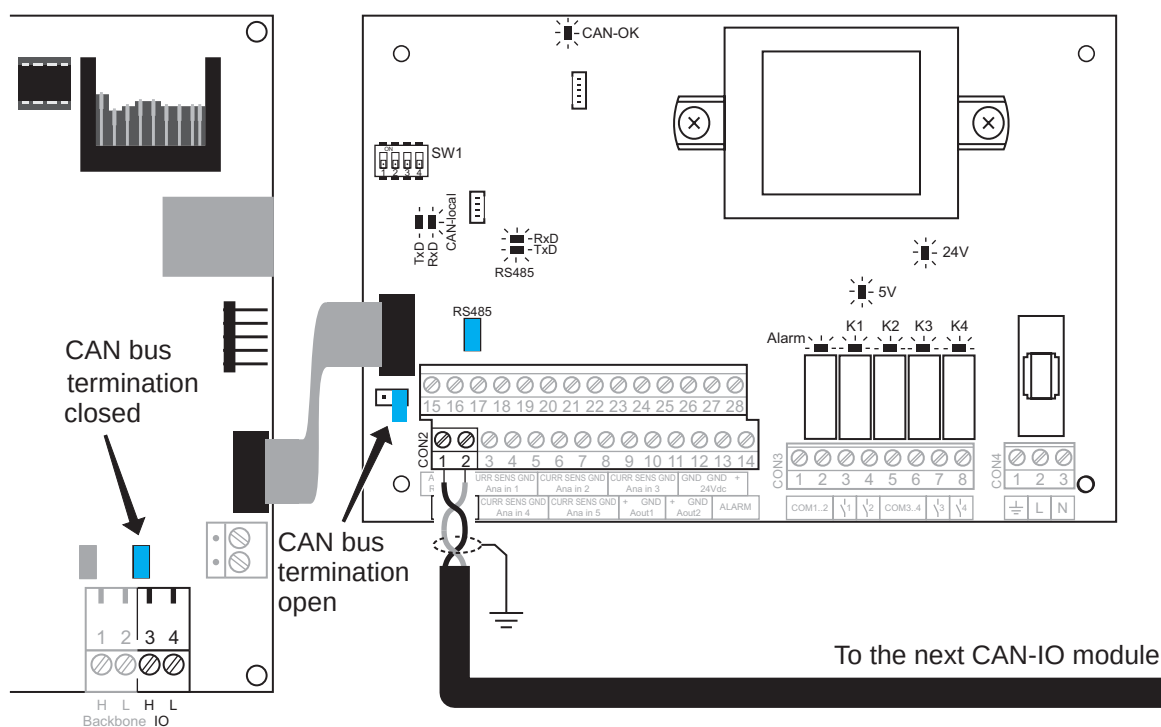
1. Module 1 - RS485 bus termination
2. Module 2 - RS485 bus termination
3. Module 3 - RS485 bus termination
4. Last module - RS485 bus termination

Note Terminate the RS485 connection with jumper JP6 at the first and last communication node.

9.4 Communication

9.4.1 CAN Local (IO modules)

The CAN Local data bus is used for communication between the Orion-MC and the IO modules (see also the examples of connections in Appendix 2).

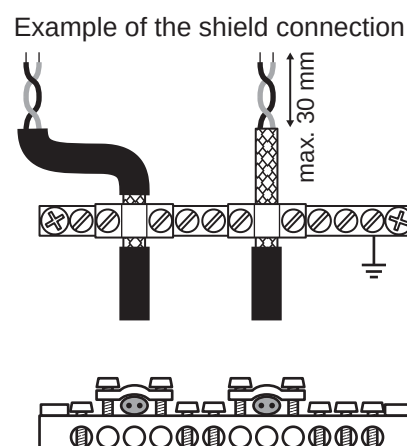


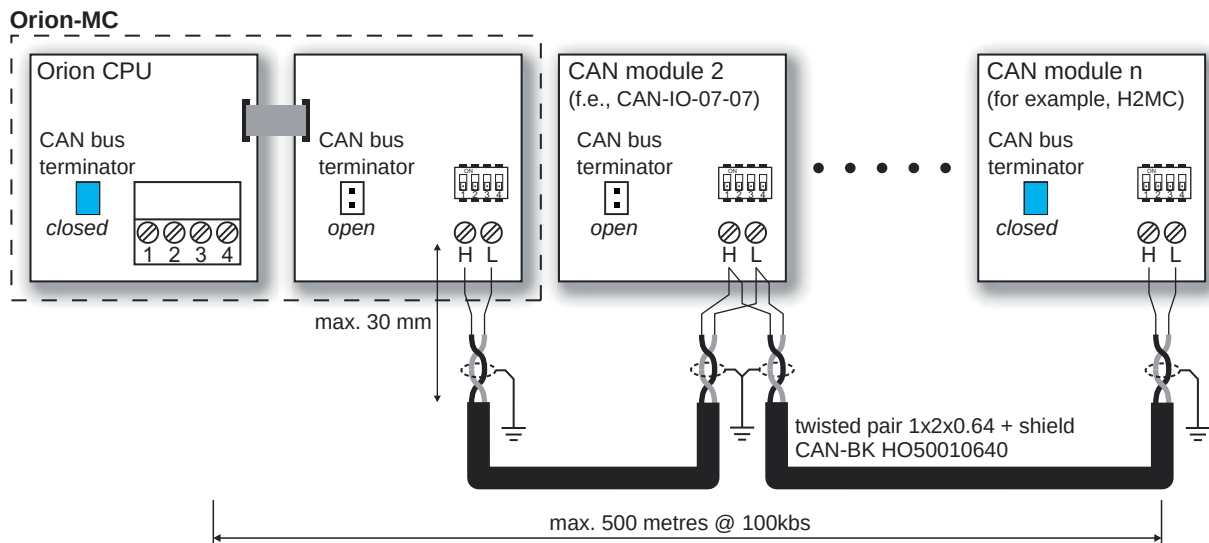
The bus must be terminated with jumpers at the first and last communication node.

Example Fit the jumpers on the CPU board and on the last IO module in the event that the CAN IO modules are connected to the integral

IO module of the Orion-MC (as shown in the example above).

(Note: remove the jumper from the integral IO module or relocate it to a position where the pins are not connected.)





Note The Orion-MC does not need to be installed at the end of the communication line.

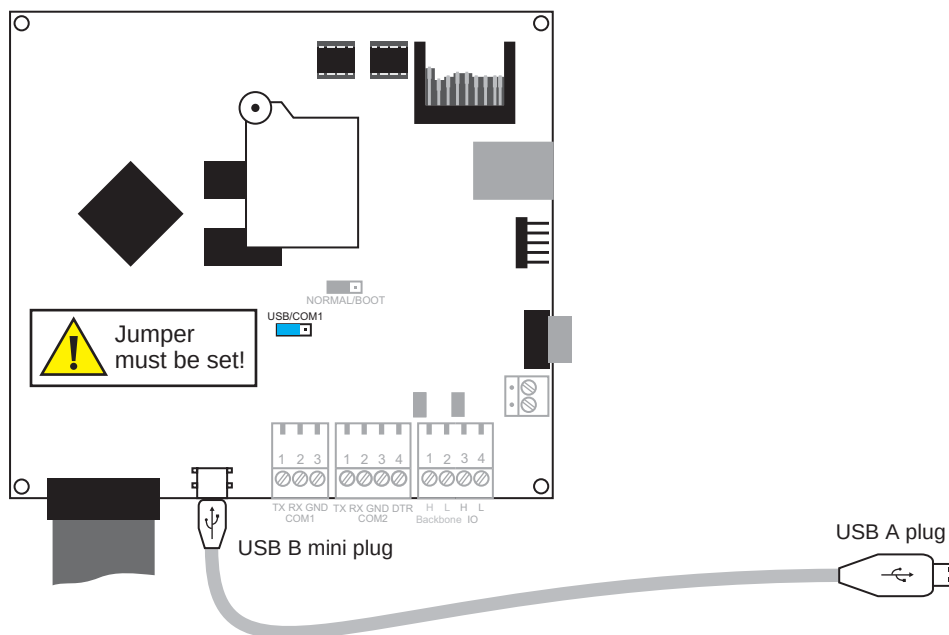
Each IO module in the communication line should be assigned a unique address. The address is configured using the IO module's DIP switch.

Address	DIP switch	Address	DIP switch	Address	DIP switch	Address	DIP switch
1		5		9		13	
2		6		10		14	
3		7		11		15	
4		8		12		16	

9.4.2 Communication with PC or Modem via USB

You can use the COM1 port or USB port (see also chapter 9.4.3). You cannot use the two ports together.

Note It is not possible to access a number of Orion-MC computers at the same time with USB.



IMPORTANT! Set the USB/COM1 jumper to USB.

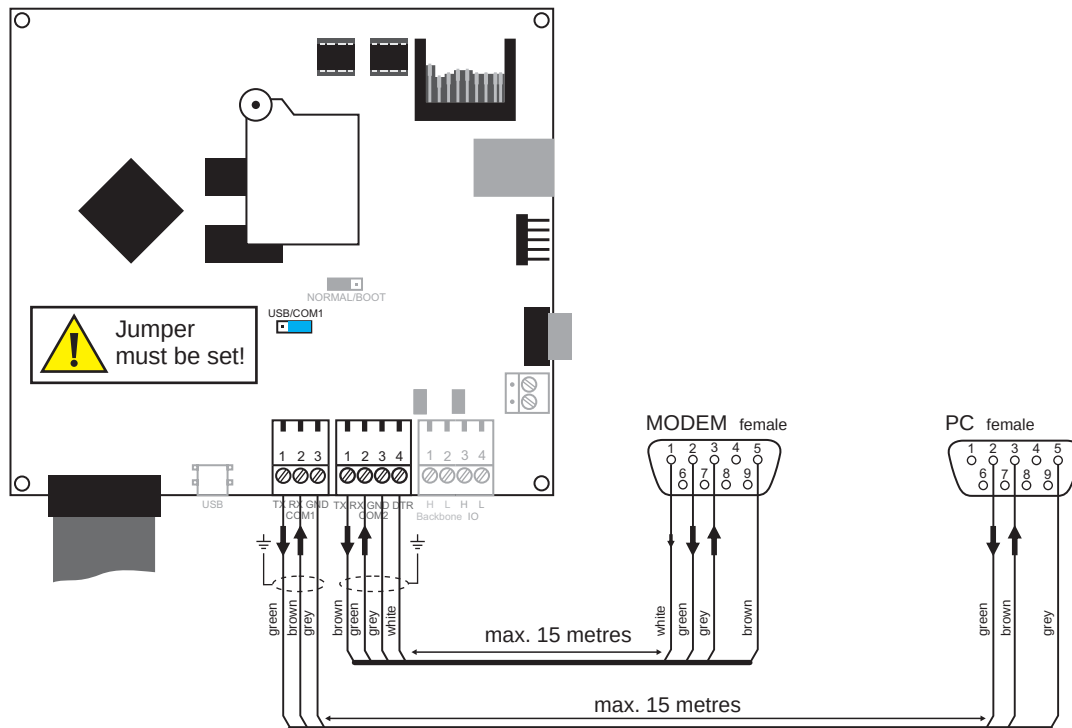
Two drivers will need to be installed for communications between the Orion-MC and a PC via USB. The general instructions for a PC with Windows are given below. Begin by installing the 'USB serial converter' driver followed by the 'USB serial port' driver.

1. Connect the Orion-MC to the PC with an USB cable.
2. Connect the Orion-MC to the power supply.
3. If Windows does not recognise the USB chip then the 'New hardware found' Wizard will start.
4. Use the Wizard to search for the drivers in the CD's 'USB' folder.
5. Install the 'USB serial converter' driver (this is selected automatically). Ignore any Windows message stating that the driver is unsigned or untested.
6. Windows will now once again find new hardware.
7. Install the 'USB serial port' driver using the same method.

9.4.3 Communication with PC or Modem via RS232

You can use the COM1 port or USB port (see also chapter 9.4.2). You cannot use the two ports together. The COM2 port can be used at any time.

Note It is not possible to access a number of Orion-MC computers at the same time with RS232.

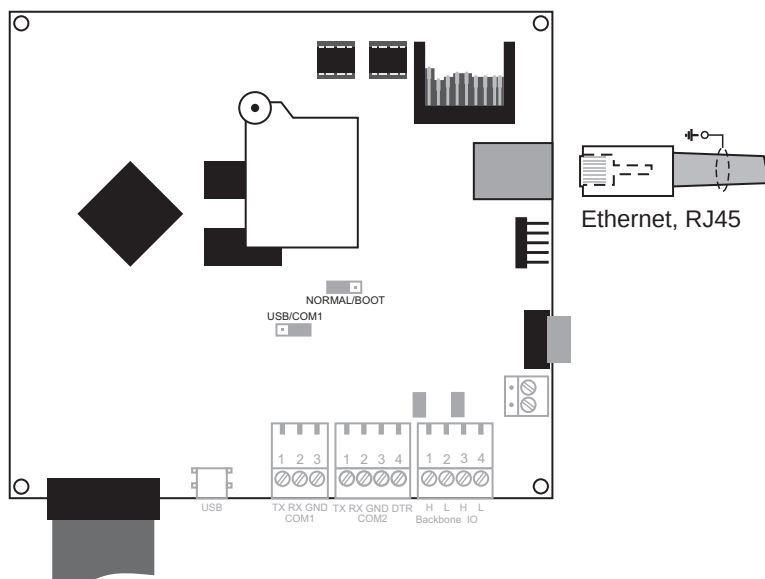


IMPORTANT! Set the USB/COM1 jumper to COM1.

Note See chapter 9.4.1 for an example of the connection of the shield.

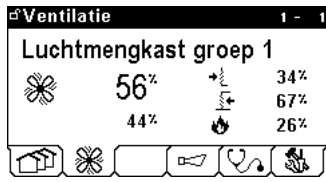
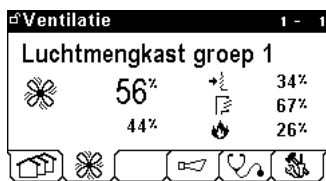
9.4.4 Communication with PC or Modem via Ethernet

You can use Ethernet to include the Orion-MC in a local IP network.



Appendix 1 Settings lists

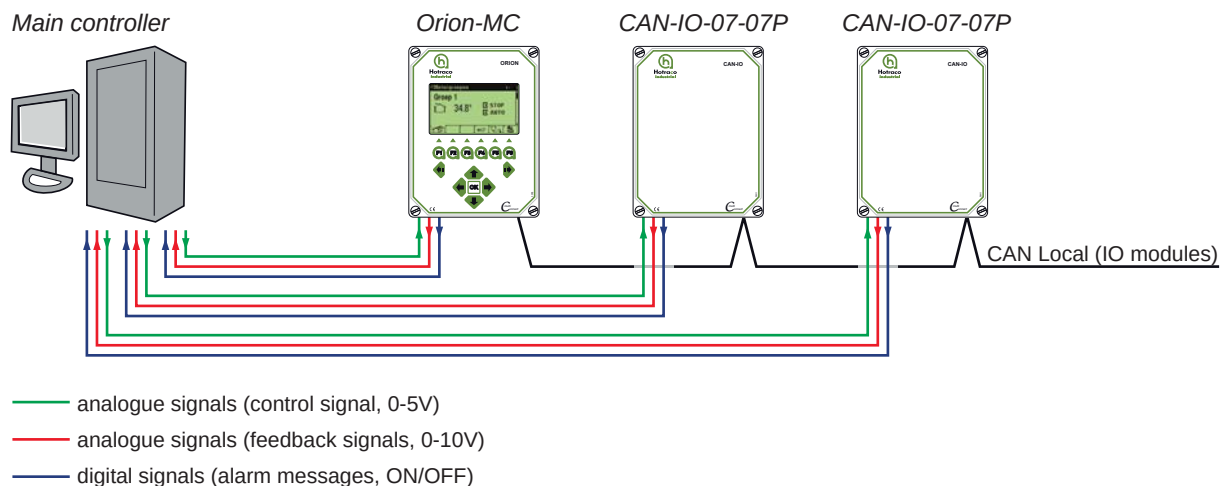
Appendix 1.1 Air mixing units

Main group	Settings	Default	User
F2  Inner flap / outer flap 	 Operation  Outer flap  Inner flap  Air blast fan	AUTO 0 % 100 % 0 %	- % % %
	Options:		
	 Air exhaust fan	0 %	%
	 Heating	0 %	%
Recirculation flap 	 Operation  Recirculation flap  Upper flap (option)  Air blast fan	AUTO 0 % 100 % 0 %	- % % %
	Options:		
	 Air exhaust fan	0 %	%
	 Heating	0 %	%

Appendix 2 System examples

Appendix 2.1 Connection to the main controller

Analog control (Orion-MC with CAN-IO-07-07P)



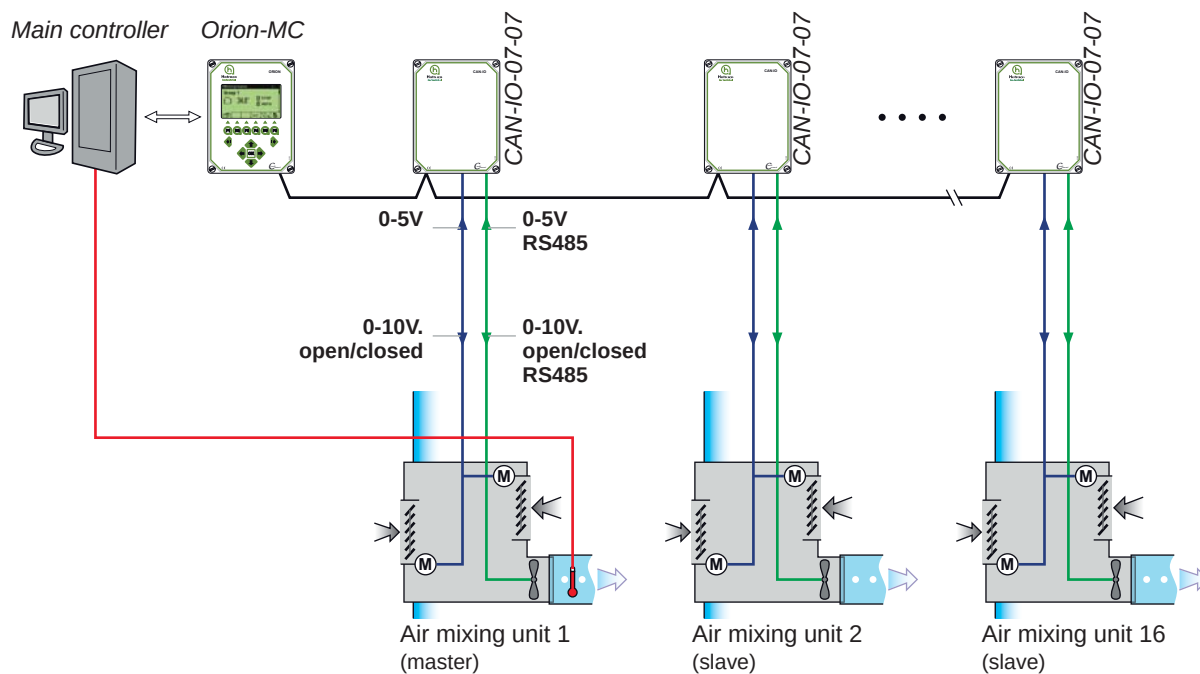
The main controller sends an analogue signal for each group to Orion-MC that corresponds with the required position. The virtual motor of the Orion-MC sends the set position as an analogue signal back to the main controller, so that the main controller receives the appropriate feedback. Digital feedback is given for alarms (the contact is de-energised in the event of an alarm).

Note

- You need one analogue input and, when relevant, one analogue output for each group: you need one digital output for each group for alarm messages.
- You can expand the number of inputs and outputs by installing extra CAN-IO-07-07P modules.

Appendix 2.2 Air mixing units

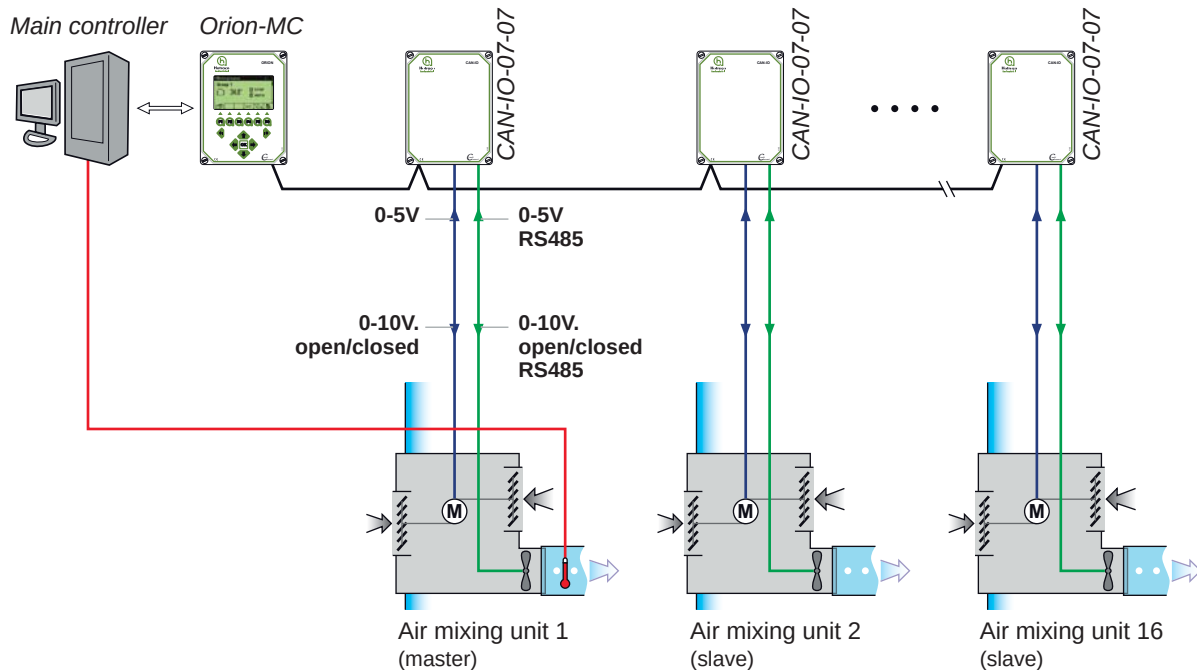
Inner/outer flap (example)



The main controller determines the position of the inner flap, outer flap and air blast fan of each group on the basis of the own measurements (usually on the basis of one air mixing unit in the group). The Orion-MC makes sure that the required positions are actually set for all air mixing units in the group.

The required positions are sent to the IO-07-07 modules via the CAN Local communication. The modules convert the information into suitable control signals for the connected air mixing unit. In addition, the air mixing unit sends the actual set positions to the Orion-MC.

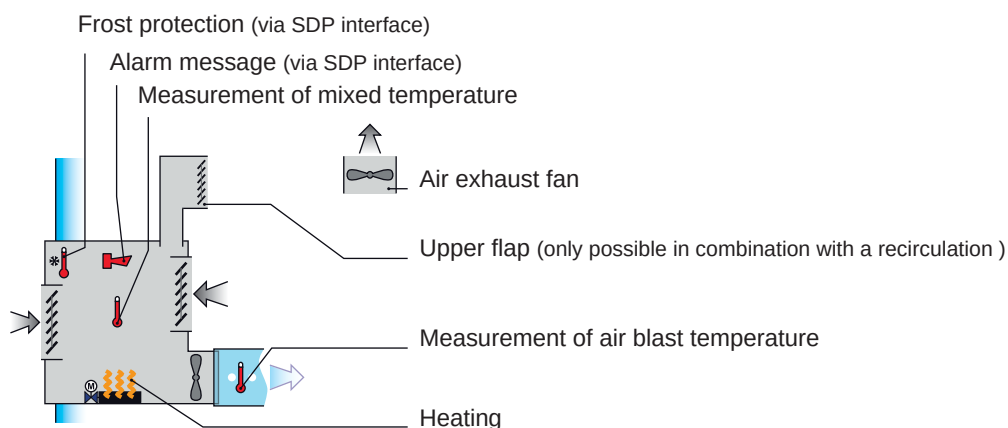
Recirculation flap (example)



The main controller determines the position of the recirculation flap and air blast fan of each group on the basis of the own measurements (usually on the basis of one air mixing unit in the group). The Orion-MC makes sure that the required positions are actually set for all air mixing units in the group.

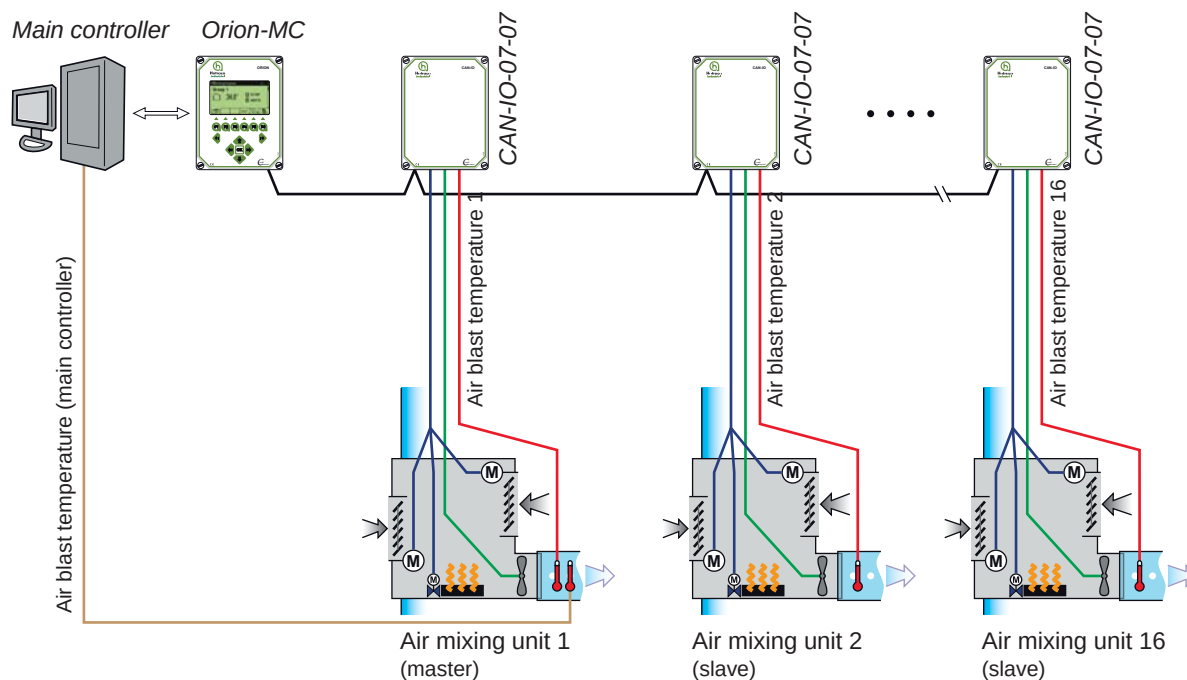
The required positions are sent to the IO-07-07 modules via the CAN Local communication. The modules convert the information into suitable control signals for the connected air mixing unit. In addition, the air mixing unit sends the actual set positions to the Orion-MC.

Options for air mixing units



Note A digital signal (alarm, thermostat) cannot be connected directly to the CAN-IO-07-07: these signals can be sent solely to a CAN-IO-07-07 equipped with an SDP interface.

Fine adjustment



The main controller determines the heating position on the basis of the own measurements of the air blast temperature of one air mixing unit (the master). The Orion-MC measures the air blast temperature of each air mixing unit (including the master).

The fine adjustment in the Orion-MC compares the air blast temperature of the slaves with the air blast temperature of the master. If necessary, the heating of a slave is increased or decreased when the air blast temperature of the slave differs from that of the master.